VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB RECORD

Artificial Intelligence (23CS5PCAIN)

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
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B.M.S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled " Bio Inspired Systems (23CS5BSBIS)" carried out by **Jeevan A(1BM22CS119),** who is bonafide student of **B.M.S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements of the above mentioned subject and the work prescribed for the said degree.

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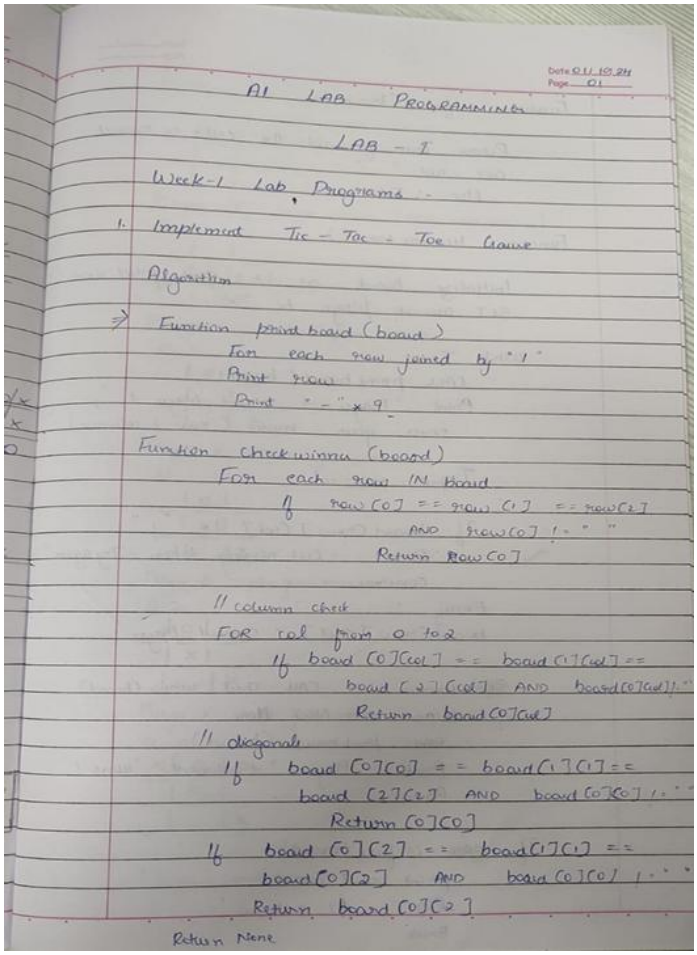
Github Link:
Jeevan-017/AI-LAB

1. Implement Tic –Tac –Toe Game.

ALGORITHM



AI LAB PROGRAMMING

LAB - 1

Week-1 Lab Programs

1. Implement Tic - Tac - Toe Game

Algorithm

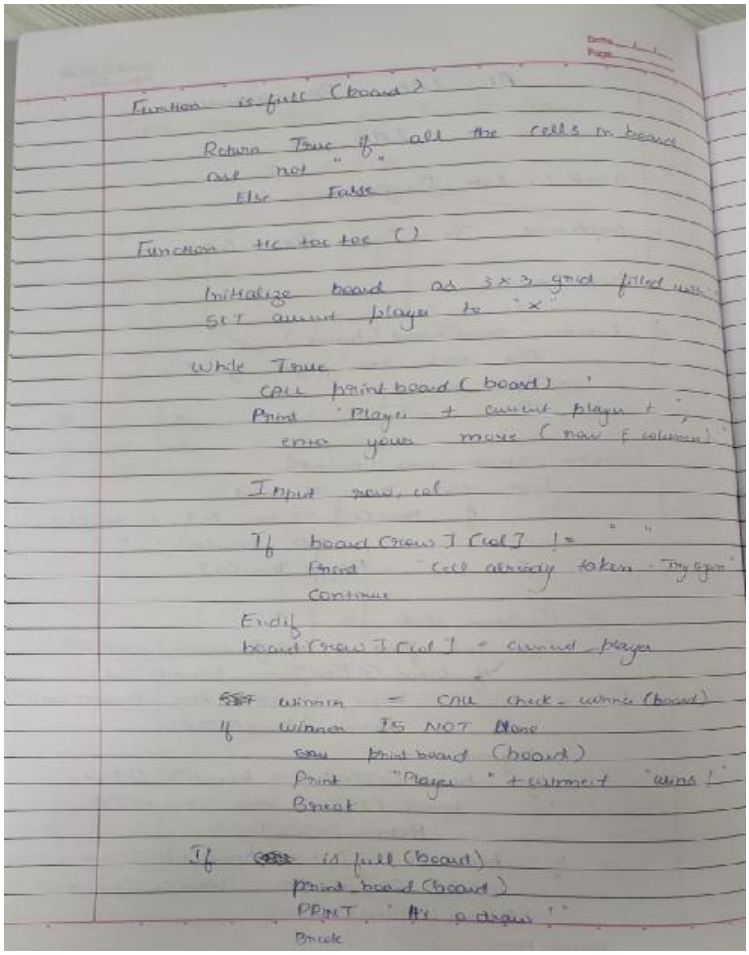
Function print board (board)
For each row joined by " | "
Print row
Print "-" x 9

Function check winner (board)
For each row IN board
If row[0] == row[1] == row[2]
AND row[0] != " "
Return Row[0]

// column check
FOR col from 0 to 2
If board[0][col] == board[1][col] ==
board[2][col] AND board[0][col] != " "
Return board[0][col]

// diagonals
If board[0][0] == board[1][1] ==
board[2][2] AND board[0][0] != " "
Return [0][0]
If board[0][2] == board[1][1] ==
board[0][2] AND board[0][0] != " "
Return board[0][2]

Return None



Function is full (board)

Return True if all the cells in board are not " "
Else False

Function tic tac toe ()

Initialize board as 3x3 grid filled with " "
SET current player to "x"

While True
CALL print board (board)
Print "Player " + current player + ", enter your move (row & column)"

Input row, col

If board[row][col] != " "
Print "Cell already taken. Try again"
Continue
Endif
board[row][col] = current player

winner = CALL check winner (board)
If winner IS NOT None
CALL print board (board)
Print "Player " + winner + " WINS!"
Break

If is full (board)
print board (board)
PRINT "It's a draw!"
Break

CODE:

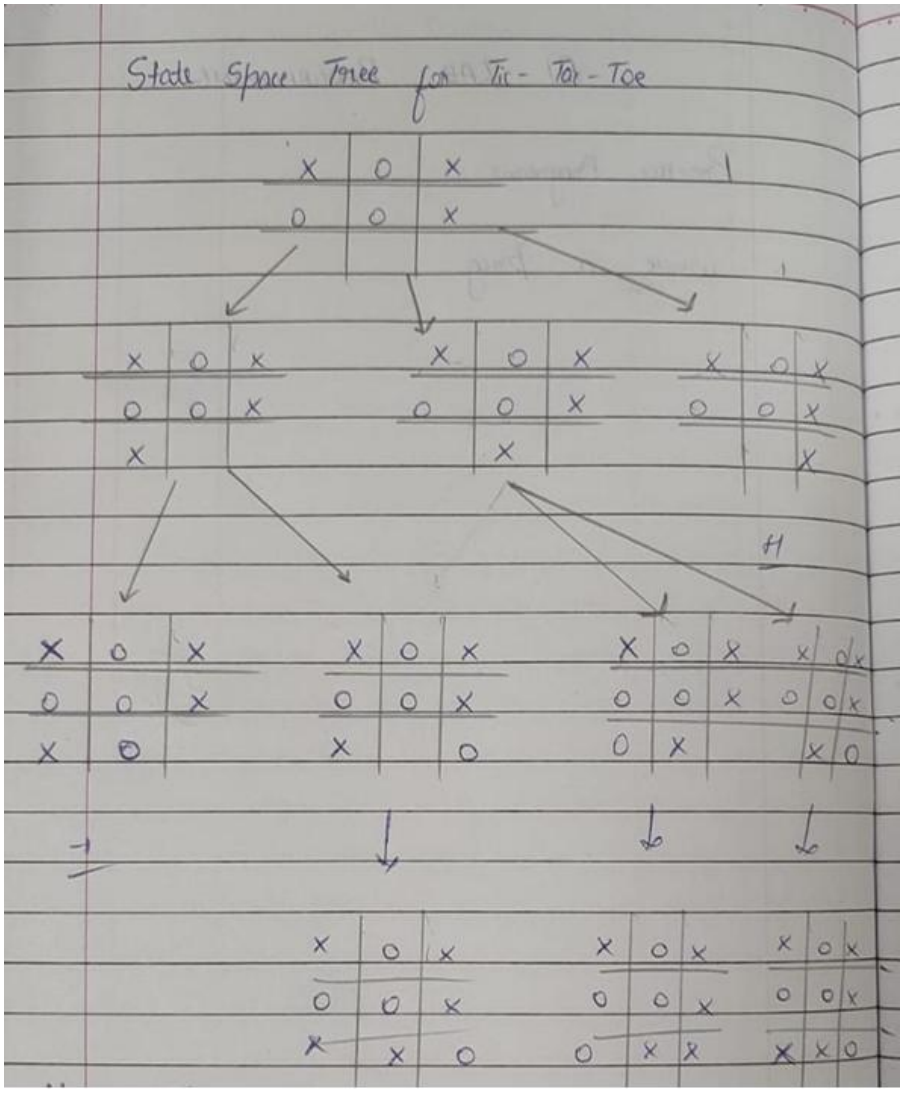
```
import random

board = ["-", "-", "-",
"-", "-", "-",
"-", "-", "-"]

def p_board(player):
print(board[0] + "|" + board[1] + "|" + board[2])
print(board[3] + "|" + board[4] + "|" + board[5])
print(board[6] + "|" + board[7] + "|" + board[8])

def check_win():

for i in range(0, 7, 3):
if board[i] == board[i + 1] == board[i + 2] and
board[i] != '-':
return board[i]

for i in range(3):
if board[i] == board[i + 3] == board[i + 6] and
board[i] != '-':
return board[i]

if board[0] == board[4] == board[8] and board[0] !=
'-':
return board[0]
if board[2] == board[4] == board[6] and board[2] !=
'-':
return board[2]
return None
def check_tie():
if '-' not in board:
return True
return False

def play_game():
current_player = "X"
game_over = False

while not game_over:
```

```
p_board(current_player)

if current_player == "X":
print("It's your turn (X).")
try:
position = int(input("Choose a position from 1-9:
")) - 1
if position < 0 or position > 8 or board[position]
!= '-':
print("Invalid move. Try again.")
continue
except ValueError:
print("Invalid input. Please enter a number
between 1 and 9.")
continue
else:
print("Computer's turn (O).")
available_moves = [i for i, spot in
enumerate(board) if spot == '-']
position = random.choice(available_moves)

board[position] = current_player
winner = check_win()

if winner:
p_board(current_player)
print(winner + " won!")
game_over = True
elif check_tie():
p_board(current_player)
print("It's a tie!")
game_over = True
else:
current_player = "O" if current_player == "X" else
"X"

play_game()
```

OUTPUT:

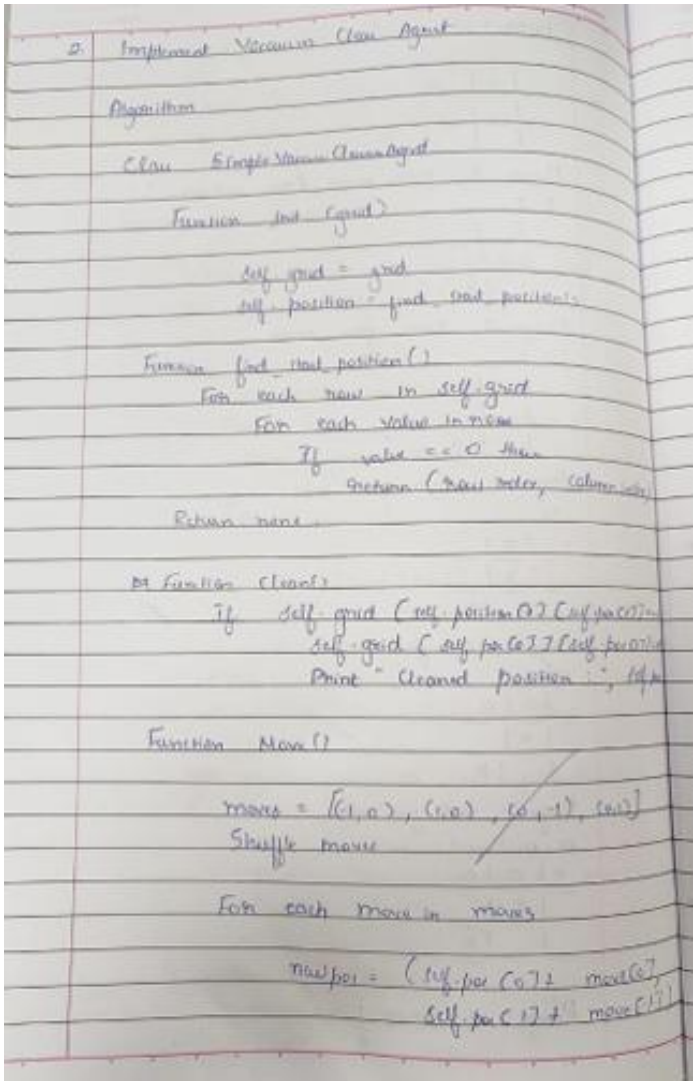
```
-|-|-
-|-|-
-|-|-
It's your turn (X).
Choose a position from 1-9: 5
-|-|-
-|X|-
-|-|-
Computer's turn (O).
-|O|-
-|X|-
-|-|-
It's your turn (X).
Choose a position from 1-9: 7
-|O|-
-|X|-
X|-|-
Computer's turn (O).
-|O|O
-|X|-
X|-|-
It's your turn (X).
Choose a position from 1-9: 1
X|O|O
-|X|-
X|-|-
Computer's turn (O).
X|O|O
-|X|O
X|-|-
It's your turn (X).
Choose a position from 1-9: 4
X|O|O
X|X|O
X|-|-
X won!
```

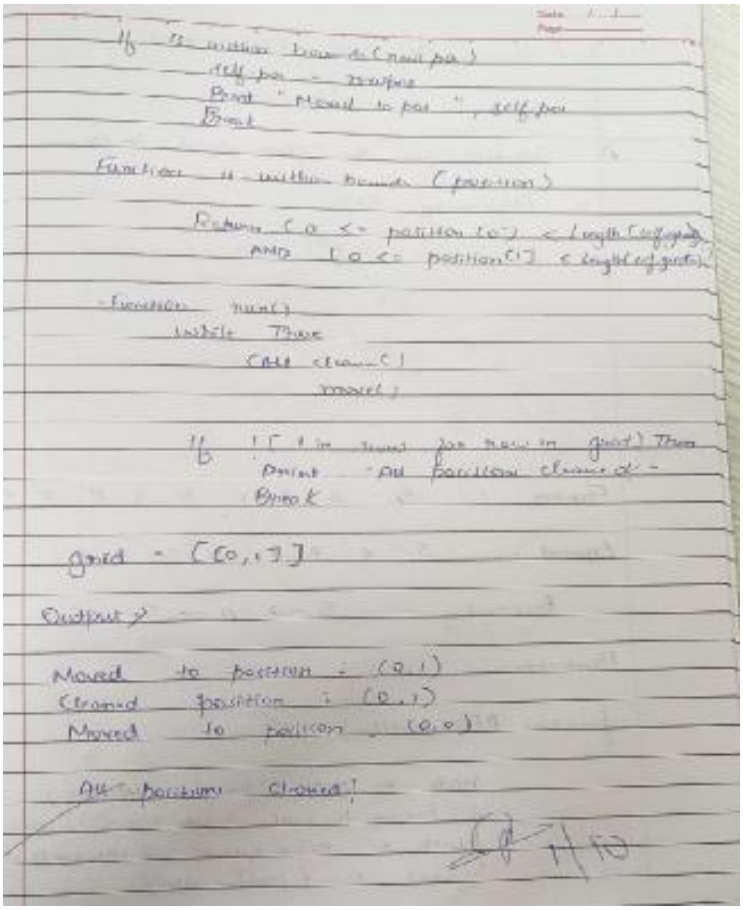
STATE-SPACE TREE:



2.Implement Vacuum Cleaner

ALGORITHM:





CODE:

```
def vacuum_world():
    goal_state = {'A': '0', 'B': '0'}
    cost = 0
    location_input = input("Enter Location of Vacuum: ")
    status_input = input("Enter status of " + location_input + " (0 for Clean, 1 for
Dirty): ")
    status_input_complement = input("Enter status of other room: ")

    print("Initial Location Condition: " + str(goal_state))

    if location_input == 'A':

        print("Vacuum is placed in Location A")

        if status_input == '1':
            print("Location A is Dirty.")

            goal_state['A'] = '0'
            cost += 1
            print("Cost for CLEANING A: " + str(cost))
            print("Location A has been Cleaned.")

            if status_input_complement == '1':
                print("Location B is Dirty.")
                print("Moving right to the Location B.")
                cost += 1
                print("COST for moving RIGHT: " + str(cost))

                goal_state['B'] = '0'
                cost += 1
                print("COST for SUCK: " + str(cost))
                print("Location B has been Cleaned.")
            else:
                print("No action. " + str(cost))
                print("Location B is already clean.")

        if status_input == '0':
```

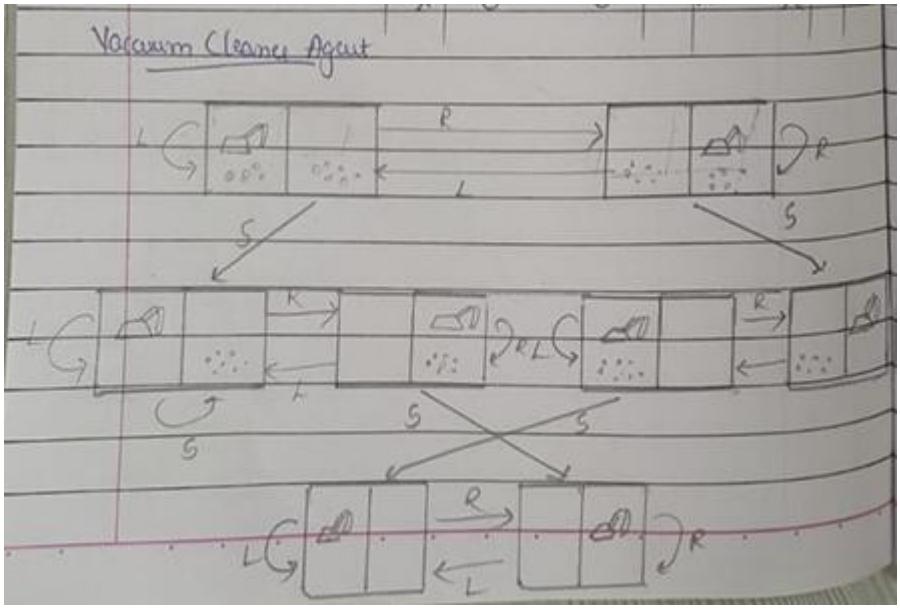
```
        print("Location A is already clean.")

        if status_input_complement == '1':
            print("Location B is Dirty.")
            print("Moving RIGHT to the Location B.")
            cost += 1
            print("COST for moving RIGHT: " + str(cost))

            goal_state['B'] = '0'
            cost += 1
            print("Cost for SUCK: " + str(cost))
            print("Location B has been Cleaned.")
        else:
            print("No action. " + str(cost))
            print("Location B is already clean.")

else:
    print("Vacuum is placed in Location B")


    if status_input == '1':
        print("Location B is Dirty.")

        goal_state['B'] = '0'
        cost += 1
        print("COST for CLEANING: " + str(cost))
        print("Location B has been Cleaned.")

        if status_input_complement == '1':
            print("Location A is Dirty.")
            print("Moving LEFT to the Location A.")
            cost += 1
            print("COST for moving LEFT: " + str(cost))

            goal_state['A'] = '0'
            cost += 1
            print("COST for SUCK: " + str(cost))
            print("Location A has been Cleaned.")
        else:
```

```
            print("Location A is already clean.")
        if status_input_complement == '1':
            print("Location A is Dirty.")
            print("Moving LEFT to the Location A.")
            cost += 1
            print("COST for moving LEFT: " + str(cost))

            goal_state['A'] = '0'
            cost += 1
            print("Cost for SUCK: " + str(cost))
            print("Location A has been Cleaned.")
        else:
            print("No action. " + str(cost))
            print("Location A is already clean.")


    print("GOAL STATE: ")
    print(goal_state)
    print("Performance Measurement: " + str(cost))


vacuum_world()
```

OUTPUT:

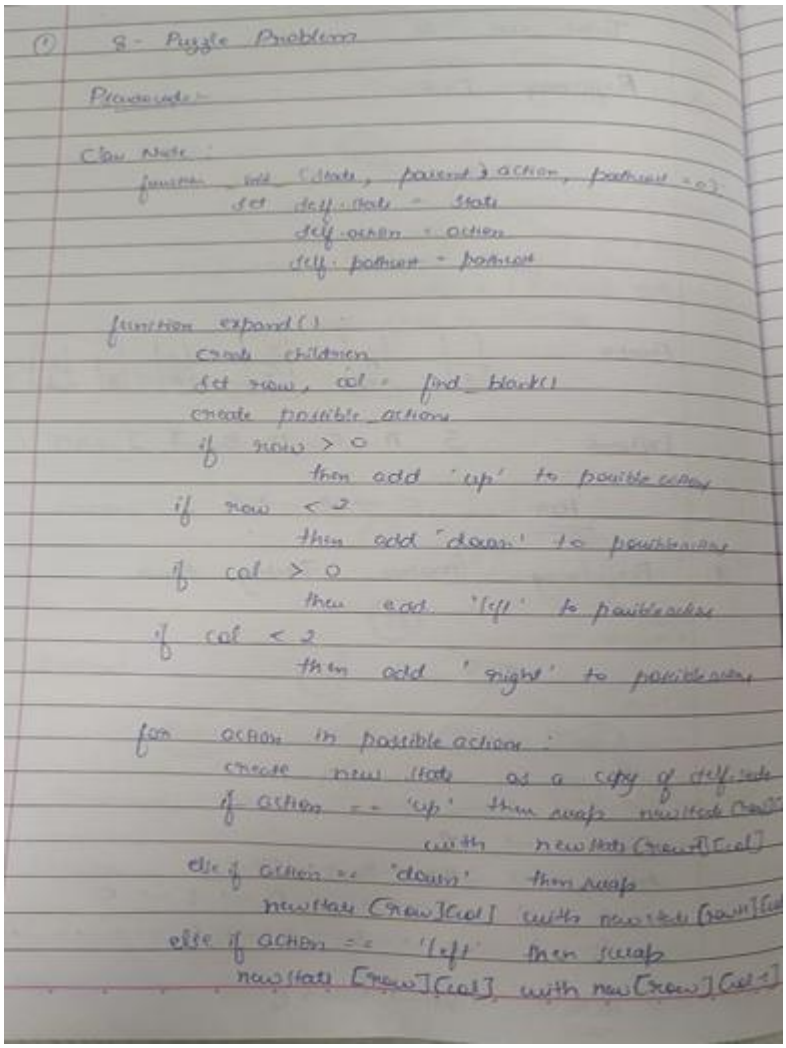
```
Enter the initial location of the vacuum cleaner (A or B): A
Enter the status of room A (0 for Clean, 1 for Dirty): 1
Enter the status of the other room (B) (0 for Clean, 1 for Dirty): 1

Initial Location Condition: {'A': '1', 'B': '1'}

Vacuum cleaner is placed in room A.
Room A is Dirty. Cleaning room A...
Cost for cleaning room A: 1

Room B is also Dirty. Moving to room B...
Cost for moving to room B: 2
Cleaning room B...
Cost for cleaning room B: 3

GOAL STATE: {'A': '0', 'B': '0'}
Total cost for cleaning: 3
```

STATE-SPACE TREE:



3.Solve 8 puzzle problems using DFS

ALGORITHM:



① 8- Puzzle Problem

Procedure:-

Class Node:
function init (state, parent, action, pathcost = 0)
set self.state = state
self.action = action
self.pathcost = pathcost

function expand()
create children
Set row, col = find_blank()
create possible_actions
if row > 0
then add 'up' to possible actions
if row < 2
then add 'down' to possible actions
if col > 0
then add 'left' to possible actions
if col < 2
then add 'right' to possible actions

for action in possible actions:
create new state as a copy of self.state
if action == 'up' then swap newState[row][col]
with newState[row-1][col]
else if action == 'down' then swap
newState[row][col] with newState[row+1][col]
else if action == 'left' then swap
newState[row][col] with new[row][col-1]

CODE:

```
class Node:
    def __init__(self, puzzle, x, y, parent=None):
        self.puzzle = [row[:] for row in puzzle]
        self.x = x
        self.y = y
        self.parent = parent

goal = [
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 0]
]

dx = [-1, 1, 0, 0]
dy = [0, 0, -1, 1]

def is_goal(puzzle):
    return puzzle == goal

def print_puzzle(puzzle):
    for row in puzzle:
        print(' '.join(str(x) if x != 0 else ' ' for x in row))
    print()

def is_valid(x, y):
    return 0 <= x < 3 and 0 <= y < 3

def count_inversions(puzzle):
    flat_puzzle = [num for row in puzzle for num in row if num != 0]
    inversions = 0
    for i in range(len(flat_puzzle)):
        for j in range(i + 1, len(flat_puzzle)):
            if flat_puzzle[i] > flat_puzzle[j]:
                inversions += 1
    return inversions

def is_solvable(puzzle):
```

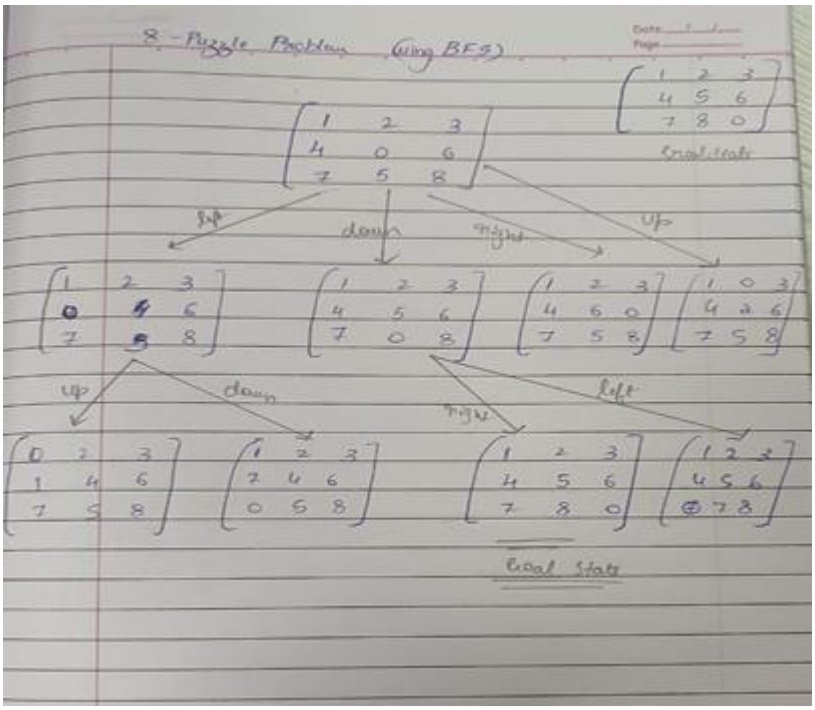
```
    return count_inversions(puzzle) % 2 == 0

def dfs(root):
    stack = [root]
    visited = set()

    while stack:
        node = stack.pop()

        if is_goal(node.puzzle):
            print("Solution found:")
            path = []
            while node:
                path.append(node.puzzle)
                node = node.parent
            for state in reversed(path):
                print_puzzle(state)
            return True

        puzzle_tuple = tuple(map(tuple, node.puzzle))
        if puzzle_tuple in visited:
            continue
        visited.add(puzzle_tuple)

        for i in range(4):
            new_x = node.x + dx[i]
            new_y = node.y + dy[i]

            if is_valid(new_x, new_y):
                new_puzzle = [row[:] for row in node.puzzle]
                new_puzzle[node.x][node.y], new_puzzle[new_x][new_y] =
new_puzzle[new_x][new_y], new_puzzle[node.x][node.y]

                new_puzzle_tuple = tuple(map(tuple, new_puzzle))
                if new_puzzle_tuple not in visited:
                    new_node = Node(new_puzzle, new_x, new_y, node)
                    stack.append(new_node)

    print("No solution found.")
```

```
    return False

def main():
    initial_puzzle = [
        [1, 2, 3],
        [4, 5, 6],
        [7, 0, 8]
    ]

    print("Initial Puzzle:")
    print_puzzle(initial_puzzle)

    if not is_solvable(initial_puzzle):
        print("The provided puzzle is unsolvable.")
        return

    try:
        x, y = [(i, j) for i in range(3) for j in range(3) if initial_puzzle[i][j] == 0][0]
    except IndexError:
        print("Invalid puzzle: No blank space (0) found.")
        return

    root = Node(initial_puzzle, x, y)

    if not dfs(root):
        print("Failed to find a solution.")

if __name__ == "__main__":
    main()
```

OUTPUT:

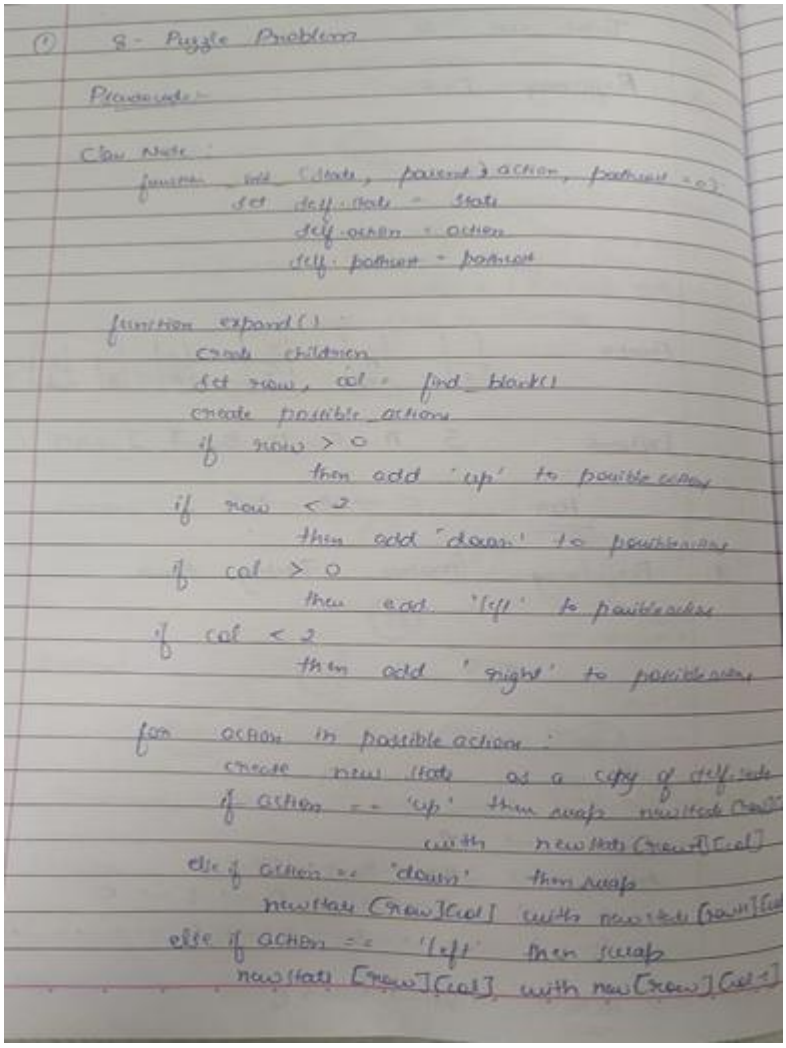
```
Initial Puzzle:
1 2 3
4 5 6
7   8

Solution found:
1 2 3
4 5 6
7   8

1 2 3
4 5 6
7 8
```

STATE-SPACE TREE:



4.Solve 8 puzzle problems using BFS

ALGORITHM:



① 8- Puzzle Problem

Pseudocode:

Class Node:
function __init__(state, parent, action, pathcost=0):
self.state = state
self.action = action
self.pathcost = pathcost

function expand():
create children
set row, col = find_blank()
create possible_actions
if row > 0
then add 'up' to possible actions
if row < 2
then add 'down' to possible actions
if col > 0
then add 'left' to possible actions
if col < 2
then add 'right' to possible actions

for action in possible actions:
create new state as a copy of self.state
if action == 'up' then swap newState[row][col]
with newState[row-1][col]
else if action == 'down' then swap
newState[row][col] with newState[row+1][col]
else if action == 'left' then swap
newState[row][col] with new[row][col-1]

CODE:

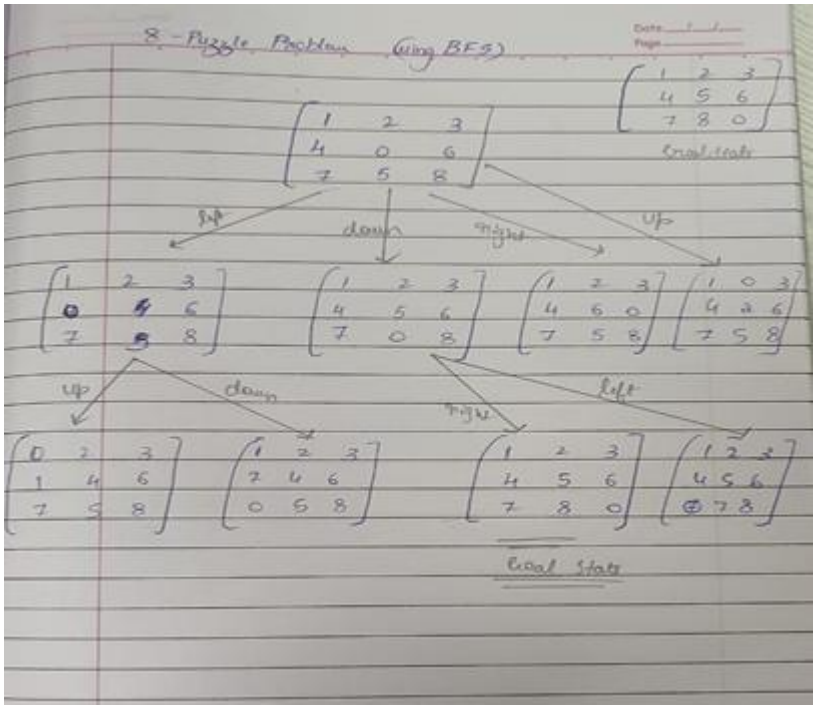
```
from collections import deque

class Node:
    def __init__(self, puzzle, x, y, parent=None):
        self.puzzle = [row[:] for row in puzzle]
        self.x = x
        self.y = y
        self.parent = parent
goal = [
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 0]
]
dx = [-1, 1, 0, 0]
dy = [0, 0, -1, 1]

def is_goal(puzzle):
    return puzzle == goal

def print_puzzle(puzzle):
    for row in puzzle:
        print(' '.join(str(x) if x != 0 else ' ' for x in row))
    print()

def is_valid(x, y):
    return 0 <= x < 3 and 0 <= y < 3

def bfs(root):
    queue = deque([root])
    visited = set()

    while queue:
        node = queue.popleft()

        if is_goal(node.puzzle):
            print("Solution found:")
            path = []
            while node:
                path.append(node.puzzle)
                node = node.parent
            for state in reversed(path):
```

```
            print_puzzle(state)
            return True

        puzzle_tuple = tuple(map(tuple, node.puzzle))
        if puzzle_tuple in visited:
            continue
        visited.add(puzzle_tuple)

        for i in range(4):
            new_x = node.x + dx[i]
            new_y = node.y + dy[i]

            if is_valid(new_x, new_y):
                new_puzzle = [row[:] for row in node.puzzle]
                new_puzzle[node.x][node.y], new_puzzle[new_x][new_y] =
new_puzzle[new_x][new_y], new_puzzle[node.x][node.y]

                new_puzzle_tuple = tuple(map(tuple, new_puzzle))
                if new_puzzle_tuple not in visited:
                    new_node = Node(new_puzzle, new_x, new_y, node)
                    queue.append(new_node)

    print("No solution found.")
    return False

def main():
    initial_puzzle = [
        [1, 2, 3],
        [4, 0, 5],
        [7, 8, 6]
    ]
    print("Initial Puzzle:")
    print_puzzle(initial_puzzle)

    x, y = [(i, j) for i in range(3) for j in range(3) if initial_puzzle[i][j] == 0][0]

    root = Node(initial_puzzle, x, y)

    if not bfs(root):
        print("Failed to find a solution.")

if __name__ == "__main__":
    main()
```

OUTPUT:

```
#Dhanush C  1BM22CS085
Initial Puzzle:
1 2 3
4   5
7 8 6

Solution found:
1 2 3
4   5
7 8 6

1 2 3
4 5
7 8 6

1 2 3
4 5 6
7 8
```

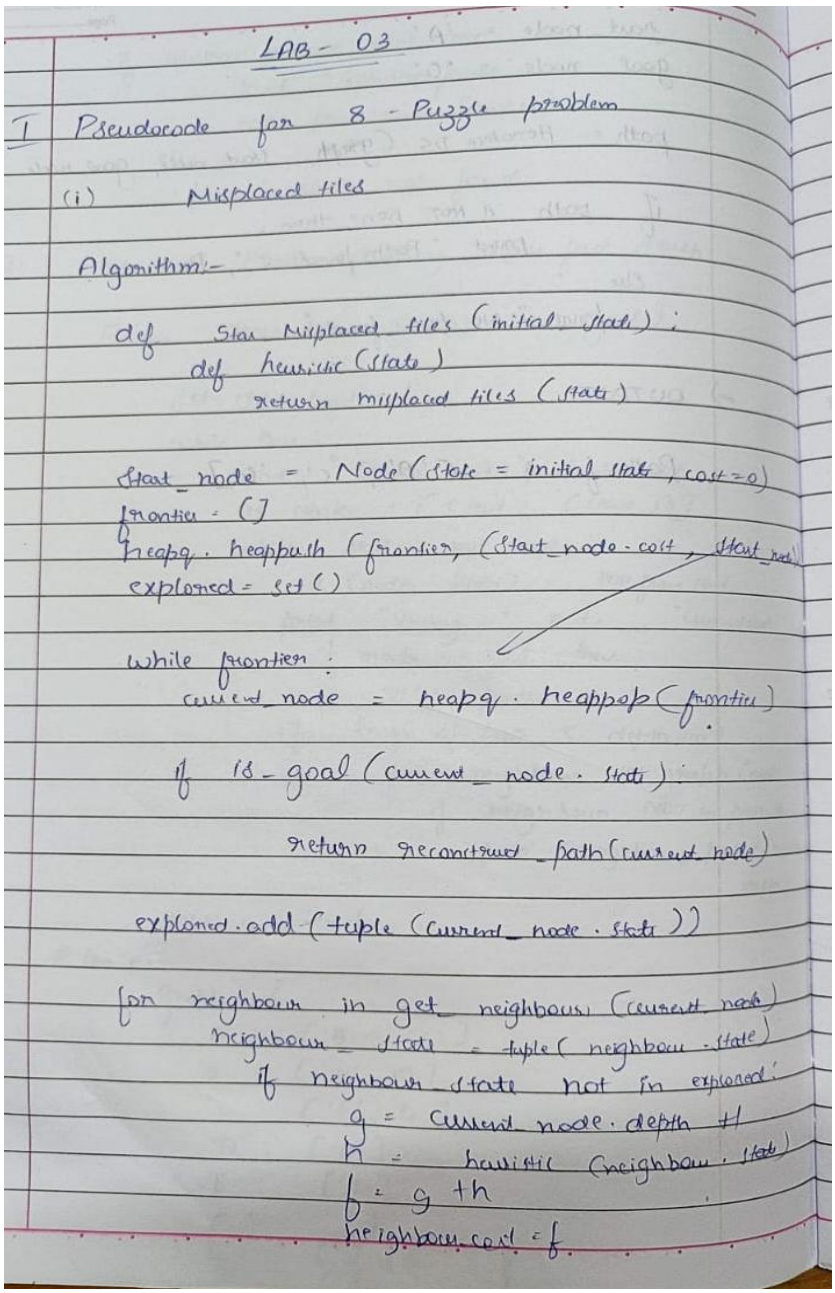
STATE-SPACE TREE:



5.Implement A* (A star) search algorithm

a.Misplacing tiles

ALGORITHM:



LAB - 03

I Pseudocode for 8 - Puzzle problem

(i) Misplaced tiles

Algorithm:-

```
def Star Misplaced tiles (initial_state):
    def heuristic (state)
        return misplaced tiles (state)

start_node = Node (state = initial state, cost=0)
frontier = []
heapq.heappush (frontier, (start_node.cost, start_node
explored = set()

while frontier:
    current_node = heapq.heappop(frontier)

    if is_goal (current_node.state):

        return reconstruct_path(current_node)

explored.add (tuple (current_node.state))

for neighbour in get_neighbours (current_node)
    neighbour_state = tuple (neighbour.state)
    if neighbour_state not in explored:
        g = current_node.depth +1
        h = heuristic (neighbour.state)
        f = g + h
        neighbour.cost = f.
```

CODE:

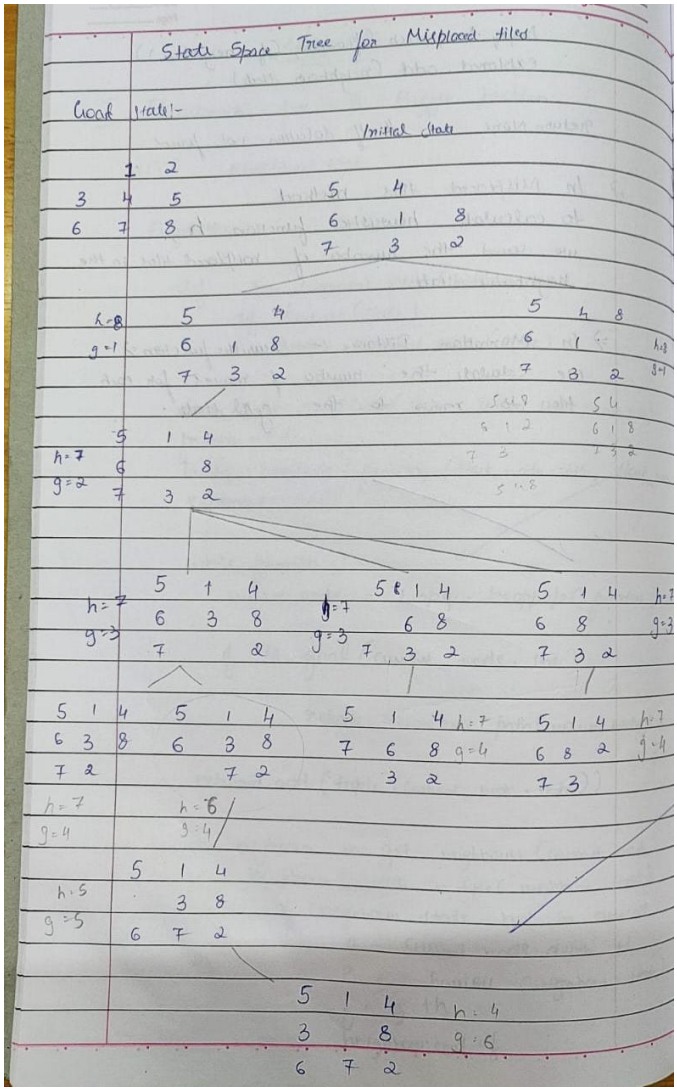
```
import heapq

def misplaced_tiles(state, goal):
    return sum(1 for i in range(3) for j in range(3)
               if state[i][j] != goal[i][j] and state[i][j] != 0)

def get_neighbors(state):
    neighbors = []
    blank = [(i, j) for i in range(3) for j in range(3) if state[i][j] == 0][0]
    possible_moves = [(-1, 0), (1, 0), (0, -1), (0, 1)]
    x, y = blank

    for dx, dy in possible_moves:
        nx, ny = x + dx, y + dy
        if 0 <= nx < 3 and 0 <= ny < 3:
            new_state = [row[:] for row in state]
            new_state[x][y], new_state[nx][ny] = new_state[nx][ny], new_state[x][y]
            neighbors.append(new_state)

    return neighbors

def print_path(path):
    for state in path:
        for row in state:
            print(row)
        print()

def astar_misplaced(start, goal):
    open_list = []
```

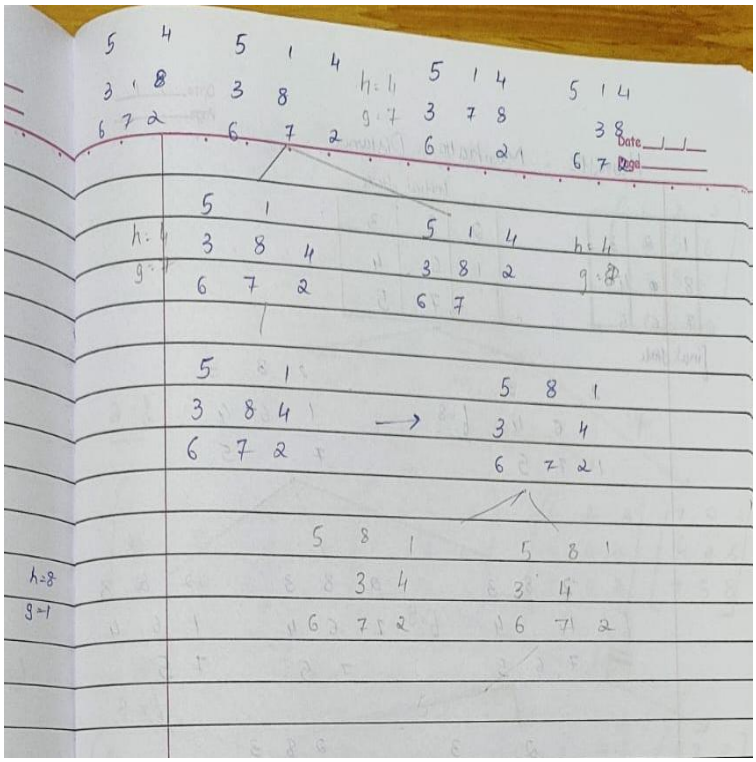
```
    heapq.heappush(open_list, (0 + misplaced_tiles(start, goal), start, 0, []))  # (f, state, g, path)
    visited = set()

    while open_list:
        f, current, g, path = heapq.heappop(open_list)
        path = path + [current]

        if current == goal:
            print("Solution Found (Misplaced Tiles):")
            print_path(path)
            return g  # Return the depth (number of moves)

        for neighbor in get_neighbors(current):
            neighbor_tuple = tuple(map(tuple, neighbor))
            if neighbor_tuple not in visited:
                visited.add(neighbor_tuple)
                h = misplaced_tiles(neighbor, goal)
                heapq.heappush(open_list, (g + 1 + h, neighbor, g + 1, path))

    return -1  # No solution found

# Example usage
start = [[5, 4, 0], [6, 1, 8], [7, 3, 2]]
goal = [[0, 1, 2], [3, 4, 5], [6, 7, 8]]
astar_misplaced(start, goal)
```

OUTPUT:

```
Solution Found (Misplaced Tiles):
[5, 4, 0]
[6, 1, 8]
[7, 3, 2]

[5, 0, 4]
[6, 1, 8]
[7, 3, 2]

[0, 5, 4]
[6, 1, 8]
[7, 3, 2]

[6, 5, 4]
[0, 1, 8]
[7, 3, 2]

[6, 5, 4]
[1, 0, 8]
[7, 3, 2]

[6, 5, 4]
[1, 3, 8]
[7, 0, 2]

[6, 5, 4]
[1, 3, 8]
[7, 2, 0]

[6, 5, 4]
[1, 3, 0]
[7, 2, 8]

[6, 5, 0]
[1, 3, 4]
[7, 2, 8]

[6, 0, 5]
[1, 3, 4]
[7, 2, 8]

[6, 3, 5]
[1, 0, 4]
[7, 2, 8]

[3, 1, 5]
[6, 0, 4]
[7, 2, 8]

[3, 1, 5]
[6, 2, 4]
[7, 0, 8]

[3, 1, 5]
[6, 2, 4]
[0, 7, 8]

[3, 1, 5]
[0, 2, 4]
[6, 7, 8]

[0, 1, 5]
[3, 2, 4]
[6, 7, 8]

[1, 0, 5]
[3, 2, 4]
[6, 7, 8]

[1, 2, 5]
[3, 0, 4]
[6, 7, 8]

[1, 2, 5]
[3, 4, 0]
[6, 7, 8]

[1, 2, 0]
[3, 4, 5]
[6, 7, 8]

[1, 0, 2]
[3, 4, 5]
[6, 7, 8]

[0, 1, 2]
[3, 4, 5]
[6, 7, 8]
```

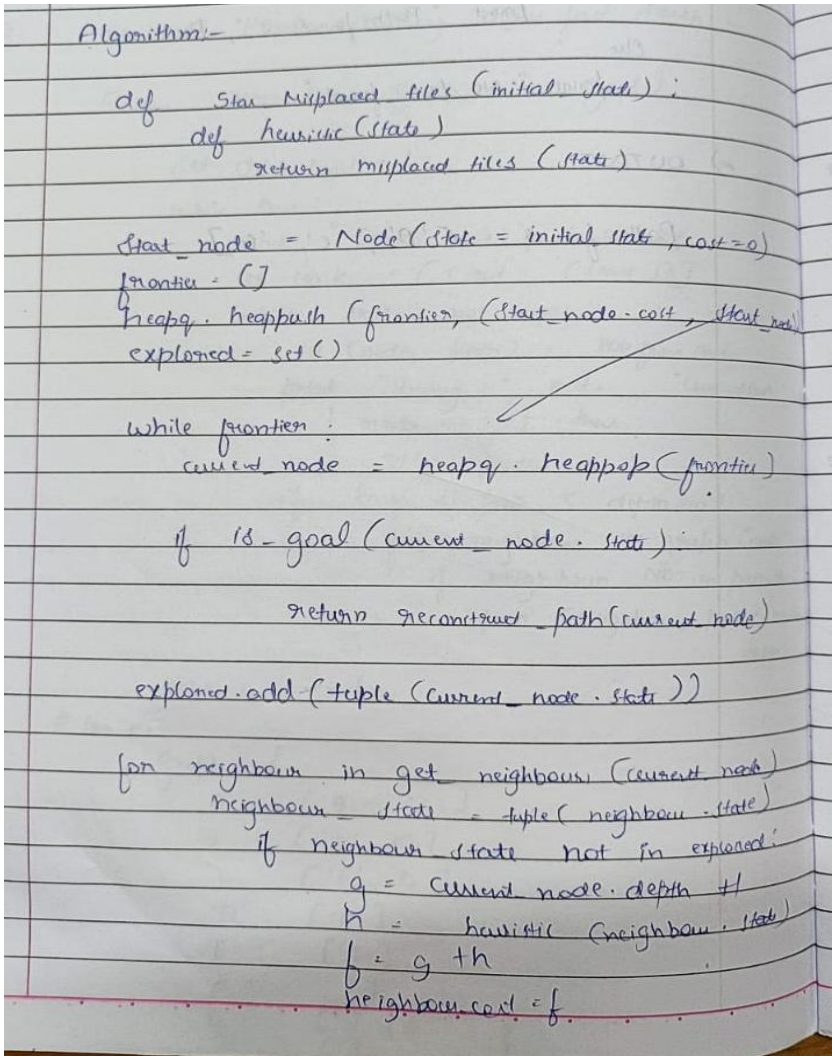
STATE-SPACE TREE:





b. Manhattan Distance

ALGORITHM:



CODE :

```
import heapq

def manhattan_distance(state, goal):
    distance = 0
```

```
    for i in range(3):
        for j in range(3):
            if state[i][j] != 0:
                goal_x, goal_y = [(x, y) for x in range(3) for y in range(3)
                                  if goal[x][y] == state[i][j]][0]
                distance += abs(i - goal_x) + abs(j - goal_y)
    return distance


def get_neighbors(state):
    neighbors = []
    blank = [(i, j) for i in range(3) for j in range(3) if state[i][j] == 0][0]
    possible_moves = [(-1, 0), (1, 0), (0, -1), (0, 1)]
    x, y = blank

    for dx, dy in possible_moves:
        nx, ny = x + dx, y + dy
        if 0 <= nx < 3 and 0 <= ny < 3:
            new_state = [row[:] for row in state]
            new_state[x][y], new_state[nx][ny] = new_state[nx][ny], new_state[x][y]
            neighbors.append(new_state)
```

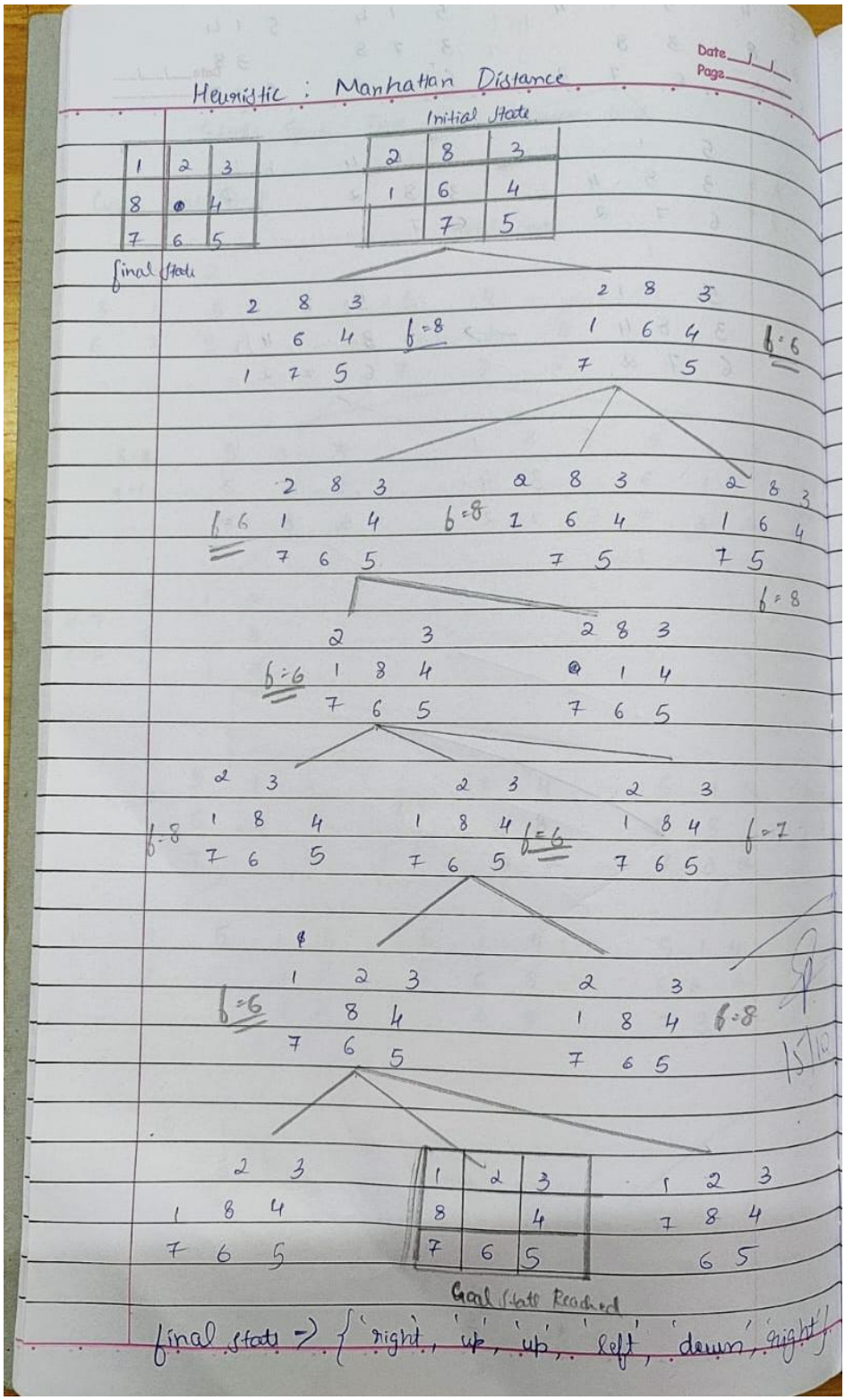
```
    return neighbors


def print_path(path):
    for state in path:
        for row in state:
            print(row)
        print()


def astar_manhattan(start, goal):
    open_list = [(start, 0, [])]  # (state, g(n), path)
    visited = set()

    while open_list:
        current, g, path = open_list.pop(0)
        path = path + [current]

        if current == goal:
            print("Solution Found (Manhattan Distance):")
            for state in path:
```

```
            distance = manhattan_distance(state, goal)
            print(f"State: {state} | Manhattan Distance: {distance}")
        print_path(path)
        return g  # Return the depth (number of moves)

    for neighbor in get_neighbors(current):
        neighbor_tuple = tuple(map(tuple, neighbor))
        if neighbor_tuple not in visited:
            visited.add(neighbor_tuple)
            h = manhattan_distance(neighbor, goal)
            open_list.append((neighbor, g + 1, path))

    open_list.sort(key=lambda x: x[1] + manhattan_distance(x[0], goal))

return -1  # No solution found

# Example usage
start = [[5, 4, 0], [6, 1, 8], [7, 3, 2]]
goal = [[0, 1, 2], [3, 4, 5], [6, 7, 8]]
astar_manhattan(start, goal)
```

OUTPUT :

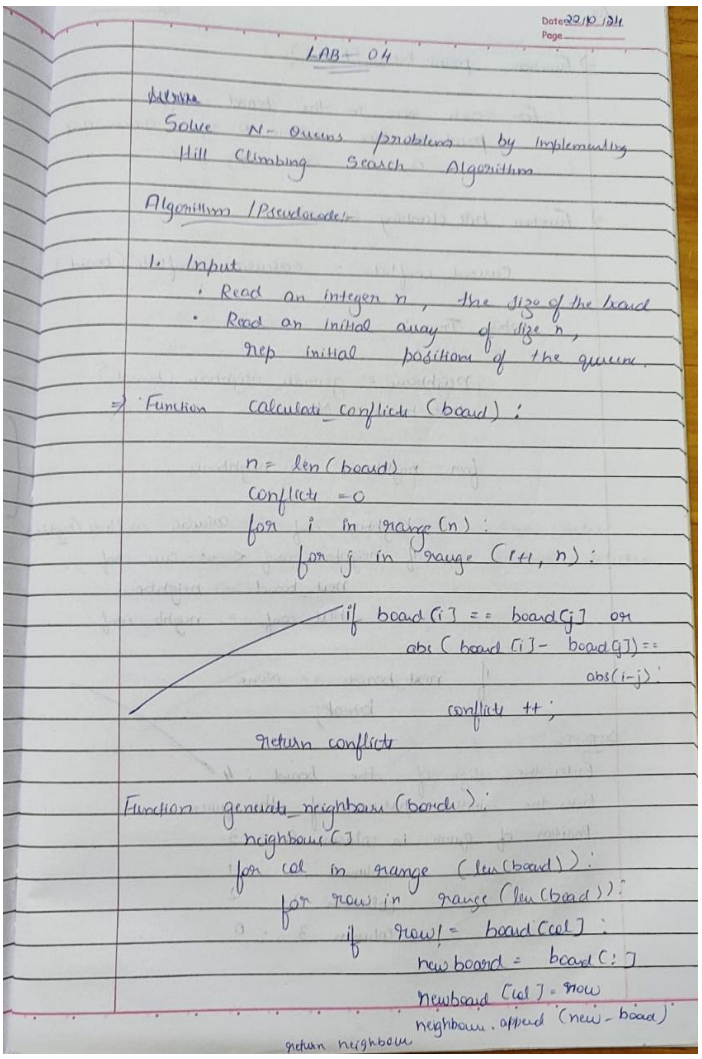
```
Solution Found (Manhattan Distance):
State: [[5, 4, 0], [6, 1, 8], [7, 3, 2]] | Manhattan Distance: 12
State: [[5, 0, 4], [6, 1, 8], [7, 3, 2]] | Manhattan Distance: 13
State: [[5, 1, 4], [6, 0, 8], [7, 3, 2]] | Manhattan Distance: 12
State: [[5, 1, 4], [6, 3, 8], [7, 0, 2]] | Manhattan Distance: 11
State: [[5, 1, 4], [6, 3, 8], [7, 2, 0]] | Manhattan Distance: 12
State: [[5, 1, 4], [6, 3, 0], [7, 2, 8]] | Manhattan Distance: 11
State: [[5, 1, 0], [6, 3, 4], [7, 2, 8]] | Manhattan Distance: 10
State: [[5, 0, 1], [6, 3, 4], [7, 2, 8]] | Manhattan Distance: 11
State: [[5, 3, 1], [6, 0, 4], [7, 2, 8]] | Manhattan Distance: 12
State: [[5, 3, 1], [6, 2, 4], [7, 0, 8]] | Manhattan Distance: 11
State: [[5, 3, 1], [6, 2, 4], [0, 7, 8]] | Manhattan Distance: 10
State: [[5, 3, 1], [0, 2, 4], [6, 7, 8]] | Manhattan Distance: 9
State: [[0, 3, 1], [5, 2, 4], [6, 7, 8]] | Manhattan Distance: 8
State: [[3, 0, 1], [5, 2, 4], [6, 7, 8]] | Manhattan Distance: 7
State: [[3, 1, 0], [5, 2, 4], [6, 7, 8]] | Manhattan Distance: 6
State: [[3, 1, 4], [5, 2, 0], [6, 7, 8]] | Manhattan Distance: 7
State: [[3, 1, 4], [5, 0, 2], [6, 7, 8]] | Manhattan Distance: 6
State: [[3, 1, 4], [0, 5, 2], [6, 7, 8]] | Manhattan Distance: 5
State: [[0, 1, 4], [3, 5, 2], [6, 7, 8]] | Manhattan Distance: 4
State: [[1, 0, 4], [3, 5, 2], [6, 7, 8]] | Manhattan Distance: 5
State: [[1, 4, 0], [3, 5, 2], [6, 7, 8]] | Manhattan Distance: 4
State: [[1, 4, 2], [3, 5, 0], [6, 7, 8]] | Manhattan Distance: 3
State: [[1, 4, 2], [3, 0, 5], [6, 7, 8]] | Manhattan Distance: 2
State: [[1, 0, 2], [3, 4, 5], [6, 7, 8]] | Manhattan Distance: 1
State: [[0, 1, 2], [3, 4, 5], [6, 7, 8]] | Manhattan Distance: 0
```

STATE SPACE TREE :



5.Implement N Queens using Hill climbing algorithm

ALGORITHM:



LAB - 04

Aim/Task
Solve N- Queens problems by implementing Hill Climbing Search Algorithm

Algorithm / Pseudocode:-

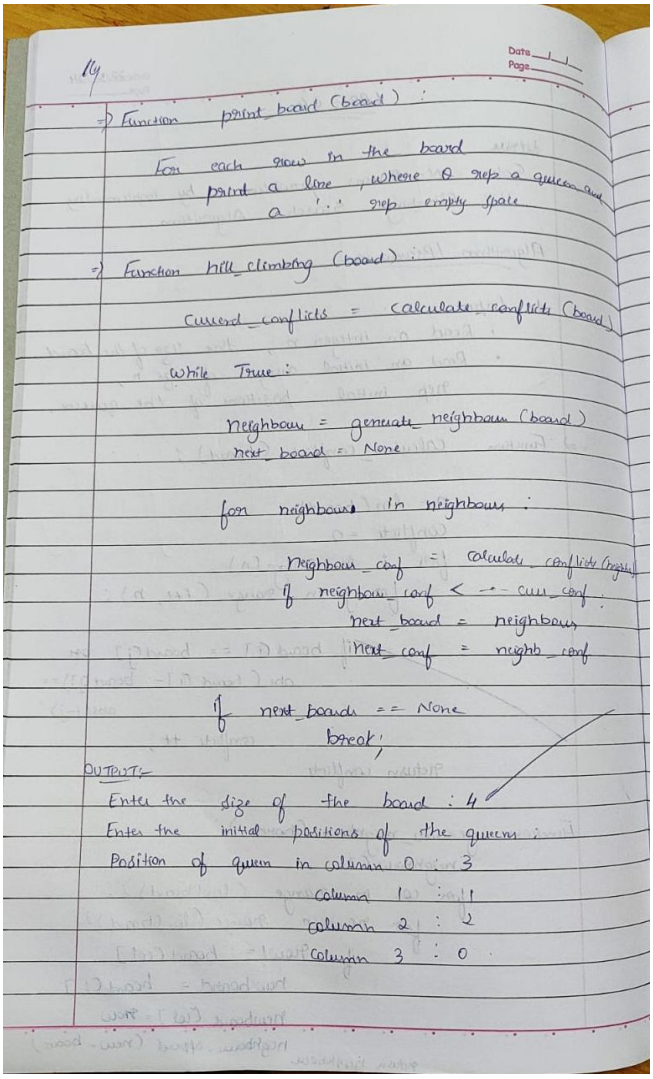
1. Input
   - Read an integer n, the size of the board
   - Read an initial array of size n, rep initial positions of the queens.

=> Function calculate_conflicts (board):

```
n = len(board)
conflicts = 0
for i in range(n):
    for j in range(i+1, n):
        if board[i] == board[j] or
           abs(board[i] - board[j]) ==
           abs(i-j):
            conflicts ++;
return conflicts
```

Function generate_neighbours (board):

```
neighbours []
for col in range (len(board)):
    for row in range (len(board)):
        if row != board[col]:
            new board = board[:]
            newboard[col] = row
            neighbours.append(new_board)
return neighbours
```



=> Function print_board (board):

For each row in the board
print a line, where Q rep a queen and a '.' rep empty space

=> Function hill_climbing (board):

```
current_conflicts = calculate_conflicts (board)
While True:
    neighbours = generate_neighbours (board)
    next_board = None
    for neighbour in neighbours:
        neighbour_conf = calculate_conflicts (neighbour)
        if neighbour_conf < current_conf:
            next_board = neighbour
            next_conf = neighb_conf
    if next_board == None
        break;
```

OUTPUT:-

```
Enter the size of the board : 4
Enter the initial positions of the queens :
Position of queen in column 0 : 3
Column 1 : 1
Column 2 : 2
Column 3 : 0
```

CODE:

```
# Function to calculate the heuristic
def calculate_heuristic(state):
    heuristic = 0
    n = len(state)
    for i in range(n):
```

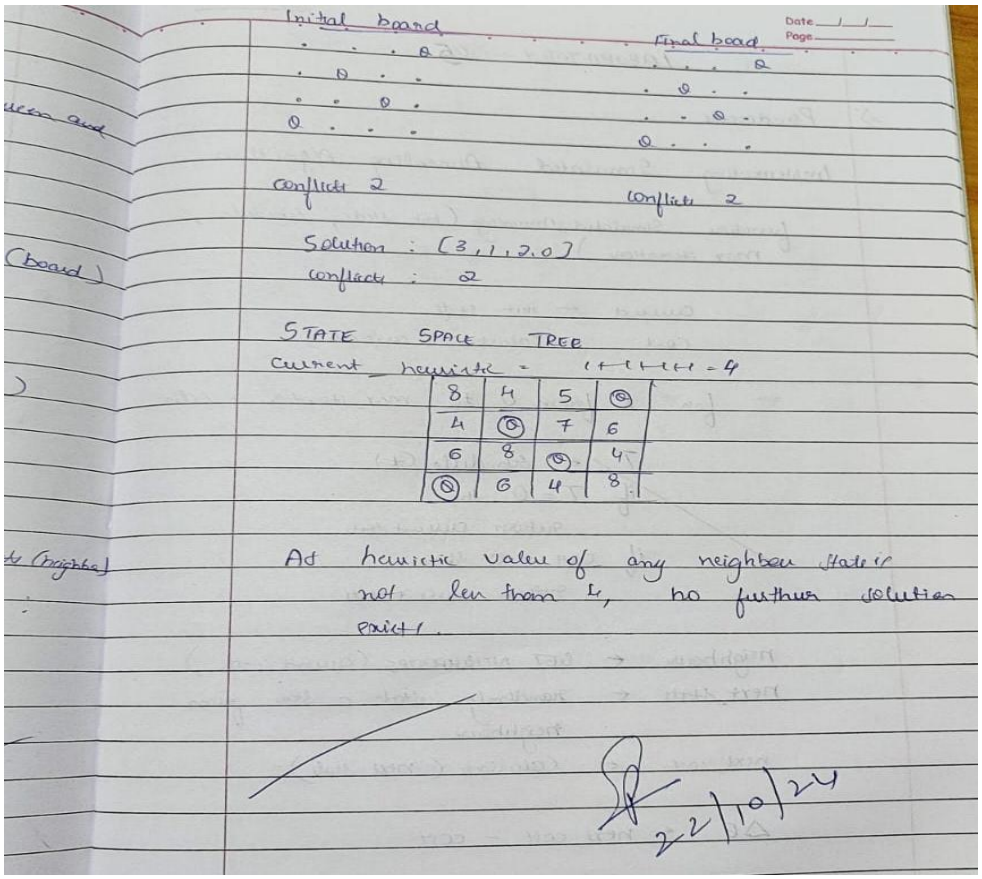
```
        for j in range(i + 1, n):
            if state[i] == state[j]:  # Same column
                heuristic += 1
            if abs(state[i] - state[j]) == abs(i - j):  # Same diagonal
                heuristic += 1
    return heuristic

# Function to generate neighboring states by swapping
def generate_neighbors(state):
    neighbors = []
    n = len(state)
    # Generate neighbors by swapping positions of two queens
    for i in range(n):
        for j in range(i + 1, n):
            new_state = state.copy()
            new_state[i], new_state[j] = new_state[j], new_state[i]  # Swap
            neighbors.append(new_state)
    return neighbors

# Function to print the board as a 2D array
def print_board(state):
    n = len(state)
    board = [['.'] * n for _ in range(n)]
    for row in range(n):
        board[row][state[row]] = 'Q'
    for row in board:
        print(' '.join(row))
    print()

# Hill Climbing algorithm for N-Queens with swapping
def hill_climbing_n_queens(initial_state):
    current_state = initial_state

    while True:
        current_heuristic = calculate_heuristic(current_state)
        print(f"Current State: {current_state}, Heuristic: {current_heuristic}")
        print_board(current_state)  # Print the current board

        if current_heuristic == 0:
```

```
            return current_state

        neighbors = generate_neighbors(current_state)
        best_neighbor = None
        best_heuristic = float('inf')

        for neighbor in neighbors:
            heuristic = calculate_heuristic(neighbor)
            if heuristic < best_heuristic:
                best_heuristic = heuristic
                best_neighbor = neighbor

        if best_heuristic >= current_heuristic:
            break  # Stop if no better neighbor found

        current_state = best_neighbor

    return None  # No solution found

# Main function to solve the N-Queens problem
def solve_n_queens():
    # Set the initial state for 4 queens
    initial_state = [3, 1, 2, 0]  # Fixed state
    solution = hill_climbing_n_queens(initial_state)

    if solution:
        print(f"Solution found for 4-Queens problem: {solution}")
        print_board(solution)  # Print the final board
    else:
        print("No solution found.")

# Run the solver
solve_n_queens()
```

OUTPUT :

```
Current State: [3, 1, 2, 0], Heuristic: 2
. . . Q
. Q . .
. . Q .
Q . . .

Current State: [1, 3, 2, 0], Heuristic: 1
. Q . .
. . . Q
. . Q .
Q . . .

Current State: [1, 3, 0, 2], Heuristic: 0
. Q . .
. . . Q
Q . . .
. . Q .

Solution found for 4-Queens problem: [1, 3, 0, 2]
. Q . .
. . . Q
Q . . .
. . Q .
```

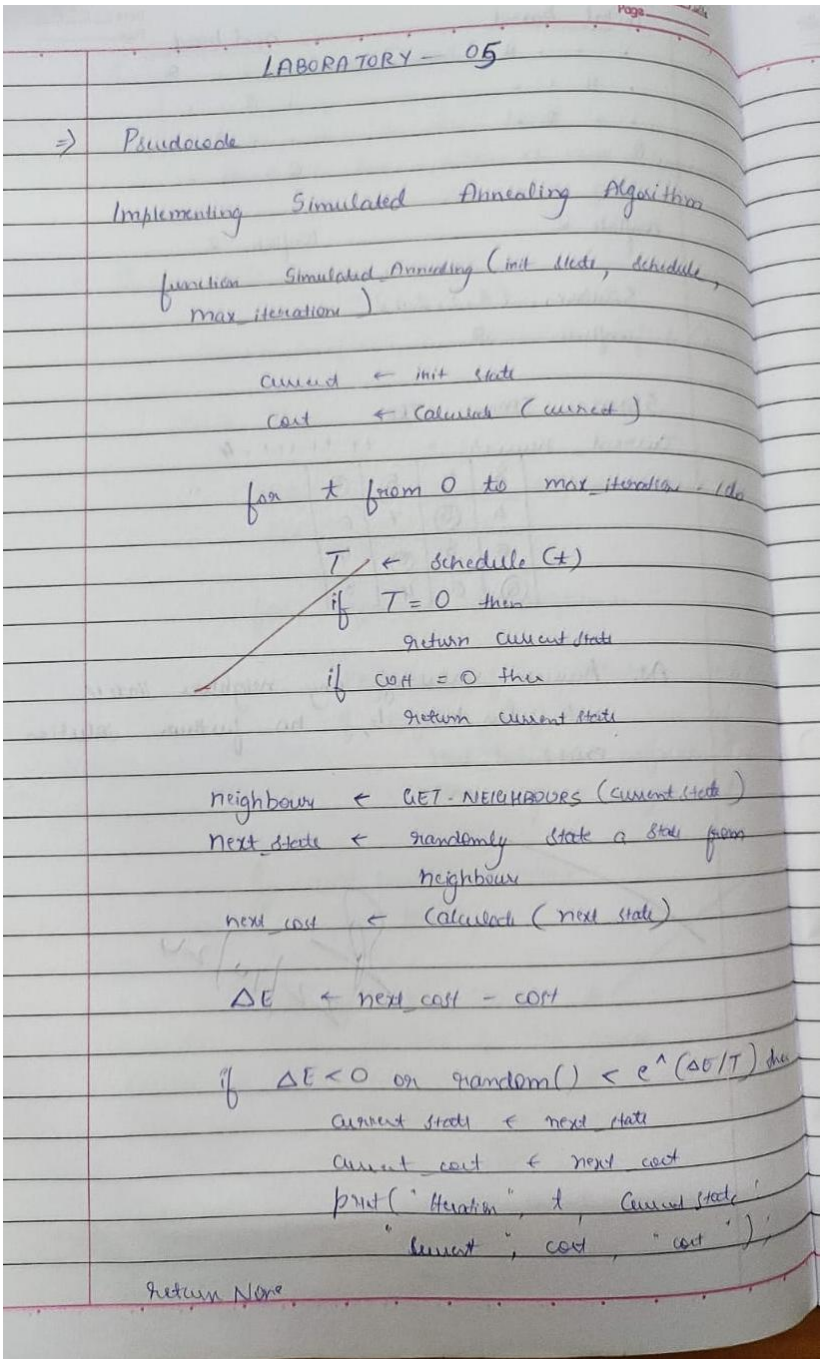
STATE-SPACE TREE:



WEEK-5

7. Simulated Annealing to solve N-Queens

ALGORITHM:



LABORATORY - 05

⇒ Pseudocode

Implementing Simulated Annealing Algorithm

function Simulated_Annealing (init state, schedule, max_iterations)

current ← init state
cost ← Calculate (current)

for t from 0 to max_iterations - 1 do

T ← schedule (t)
if T = 0 then
return current state
if cost = 0 then
return current state

neighbour ← GET-NEIGHBOURS (current state)
next_state ← randomly state a state from neighbour
next_cost ← Calculate (next state)

ΔE ← next_cost - cost

if ΔE < 0 or random() < e^(ΔE/T) then
current state ← next_state
current_cost ← next_cost
print ("Iteration", t, "current state", "current", cost, "cost");

return None

CODE:

```
import random
import math

def create_board(n, initial_config=None):
    """Create a board with the specified initial configuration."""
    if initial_config:
        return initial_config  # Use the provided initial configuration
    return [random.randint(0, n-1) for _ in range(n)]

def count_conflicts(board):
    """Count the number of pairs of queens that are attacking each other."""
    n = len(board)
    conflicts = 0
    for i in range(n):
        for j in range(i+1, n):
            if board[i] == board[j] or abs(board[i] - board[j]) == j - i:
                conflicts += 1
    return conflicts

def get_neighbors(board):
    """Generate neighbors by moving one queen at a time to a different row."""
    neighbors = []
    for i in range(len(board)):
        for row in range(len(board)):
            if row != board[i]:
                neighbor = board[:]
                neighbor[i] = row
                neighbors.append(neighbor)
    return neighbors

def print_board(board):
    """Print the board in a human-readable format."""
    n = len(board)
    for row in range(n):
        line = ['Q' if board[col] == row else '.' for col in range(n)]
        print(' '.join(line))
    print()
```

```
def simulated_annealing(n, initial_config, max_iterations=1000,
initial_temperature=100, cooling_rate=0.95):
    """Solve the N-Queens problem using simulated annealing."""
    current_board = create_board(n, initial_config)
    current_conflicts = count_conflicts(current_board)
    temperature = initial_temperature

    best_board = current_board[:]
    best_conflicts = current_conflicts

    # Print the initial configuration
    print("Initial configuration:")
    print_board(current_board)
    print(f"Initial number of conflicts: {current_conflicts}\n")

    for iteration in range(max_iterations):
        if current_conflicts == 0:
            # If the solution is found, print the final configuration and exit
            print(f"Solution found at iteration {iteration + 1}!")
            print("Final configuration (solution):")
            print_board(current_board)
            return current_board

        # Get neighbors of the current board
        neighbors = get_neighbors(current_board)
        next_board = random.choice(neighbors)
        next_conflicts = count_conflicts(next_board)

        # If the neighbor has fewer conflicts, accept it
        if next_conflicts < current_conflicts:
            current_board = next_board
            current_conflicts = next_conflicts
        else:
            # Accept the neighbor with a certain probability
            probability = math.exp((current_conflicts - next_conflicts) / temperature)
            if random.random() < probability:
                current_board = next_board
                current_conflicts = next_conflicts
```

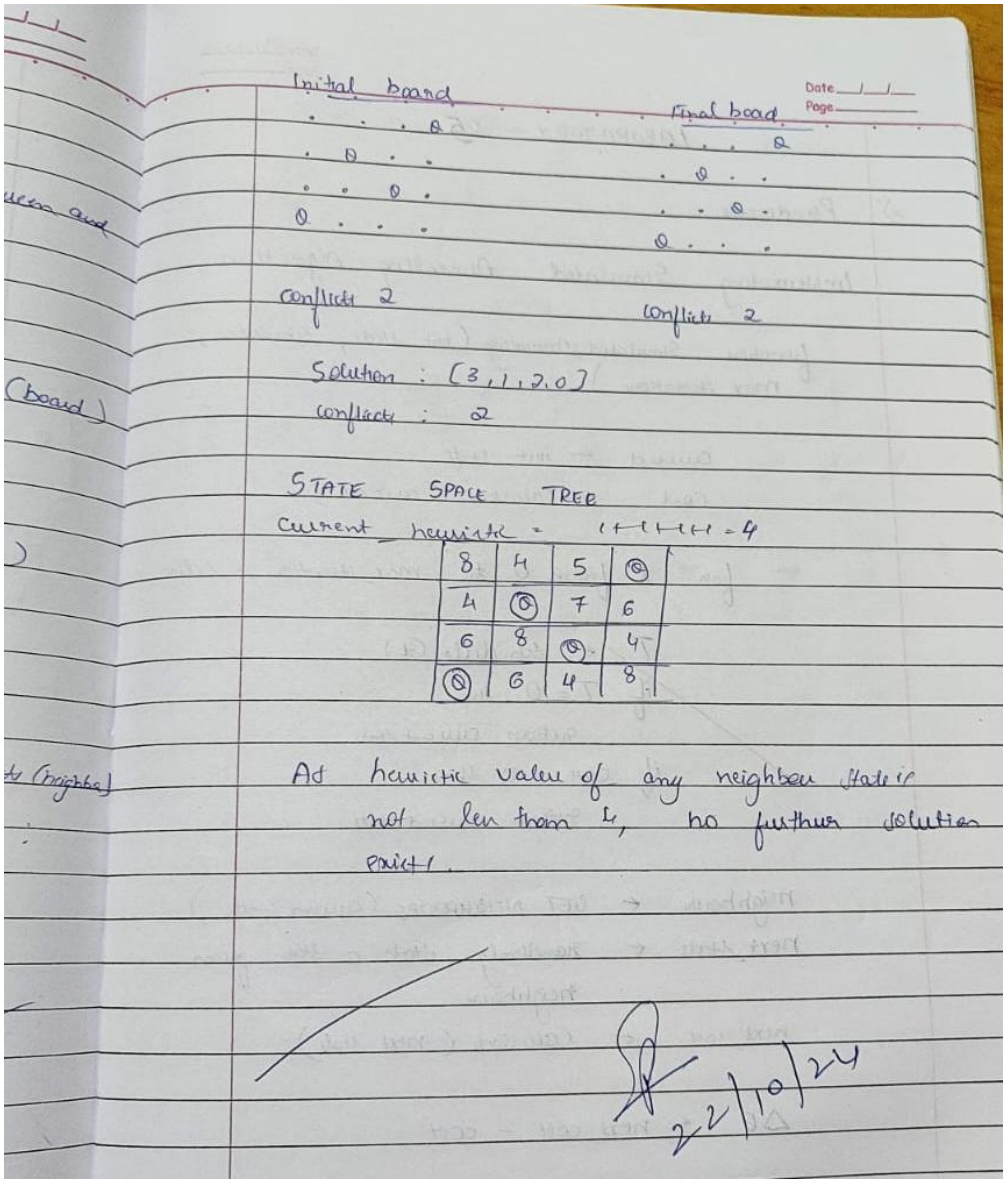
```
        # Update temperature
        temperature *= cooling_rate

    # If no solution is found, print the best state after max iterations
    print(f"No solution found within {max_iterations} iterations.")
    print("Best configuration found:")
    print_board(best_board)
    return best_board

# Set the number of queens (4 for this example) and initial configuration
n = 4
initial_config = [2, 1, 3, 0]  # The custom configuration
solution = simulated_annealing(n, initial_config)
```

OUTPUT:

Output

```
Initial configuration:
. . . Q
. Q . .
Q . . .
. . Q .

Initial number of conflicts: 1

Solution found at iteration 3!
Final configuration (solution):
. Q . .
. . . Q
Q . . .
. . Q .

=== Code Execution Successful ===
```

STATE SPACE TREE:



Initial board

. . . Q
. Q . .
. . Q .
Q . . .

Conflicts 2

Final board

. . . Q
. Q . .
. . Q .
Q . . .

Conflicts 2

Solution : [3, 1, 2, 0]
conflicts : 2

STATE SPACE TREE

Current heuristic = 1+1+1+1 = 4

8	4	5	(3)
4	(1)	7	6
6	8	(2)	4
(3)	6	4	8

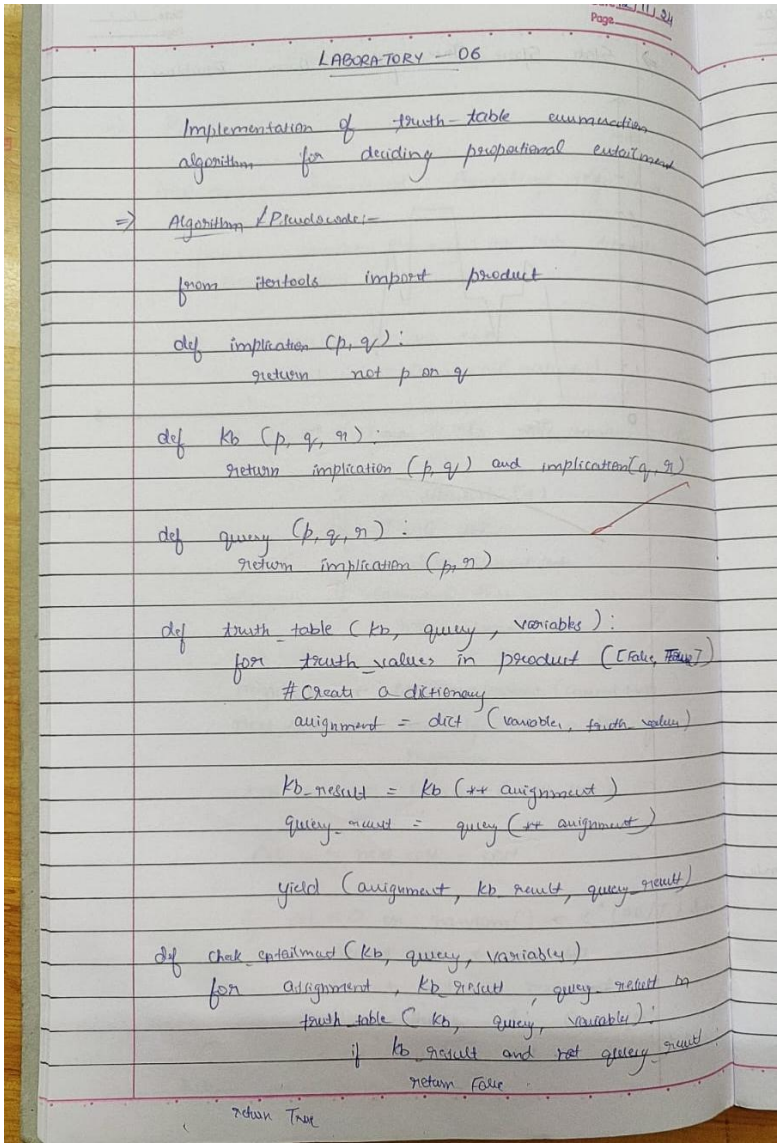
At heuristic value of any neighbour state is not less than 4, no further solution exists.

22/10/24

WEEK-6

8. Propositional Logic

ALGORITHM:



LABORATORY – 06

Implementation of truth-table enumeration algorithm for deciding propositional entailment

=> Algorithm / Pseudocode:-

```
from itertools import product

def implication (p, q):
    return not p or q

def kb (p, q, r):
    return implication (p, q) and implication (q, r)

def query (p, q, r):
    return implication (p, r)

def truth_table (kb, query, variables):
    for truth_values in product ([False, True])
        # Create a dictionary
        assignment = dict (variable, truth_values)

        kb_result = kb (** assignment)
        query_result = query (** assignment)

        yield (assignment, kb_result, query_result)

def check_entailment (kb, query, variables)
    for assignment, kb_result, query_result in
        truth_table (kb, query, variables):
            if kb_result and not query_result:
                return False
    return True
```

CODE:

```
combinations = [(True, True, True), (True, True, False), (True, False, True), (True, False, False),
                (False, True, True), (False, True, False), (False, False, True), (False, False, False)]
variable = {'p': 0, 'q': 1, 'r': 2}
kb = ''
q = ''
priority = {'~': 3, 'v': 1, '^': 2}

def input_rules():
    global kb, q
    kb = input("Enter rule: ")
    q = input("Enter the Query: ")

def entailment():
    global kb, q
    print('*' * 10 + "Truth Table Reference" + '*' * 10)
    print('kb', 'alpha')
    print('*' * 10)
    for comb in combinations:
        s = evaluatePostfix(toPostfix(kb), comb)
        f = evaluatePostfix(toPostfix(q), comb)
        print(s, f)
        print('-' * 10)
        if s and not f:
            return False
    return True

def isOperand(c):
    return c.isalpha() and c != 'v'

def isLeftParanthesis(c):
    return c == '('

def isRightParanthesis(c):
    return c == ')'

def isEmpty(stack):
    return len(stack) == 0

def peek(stack):
    return stack[-1]

def hasLessOrEqualPriority(c1, c2):
    try:
```

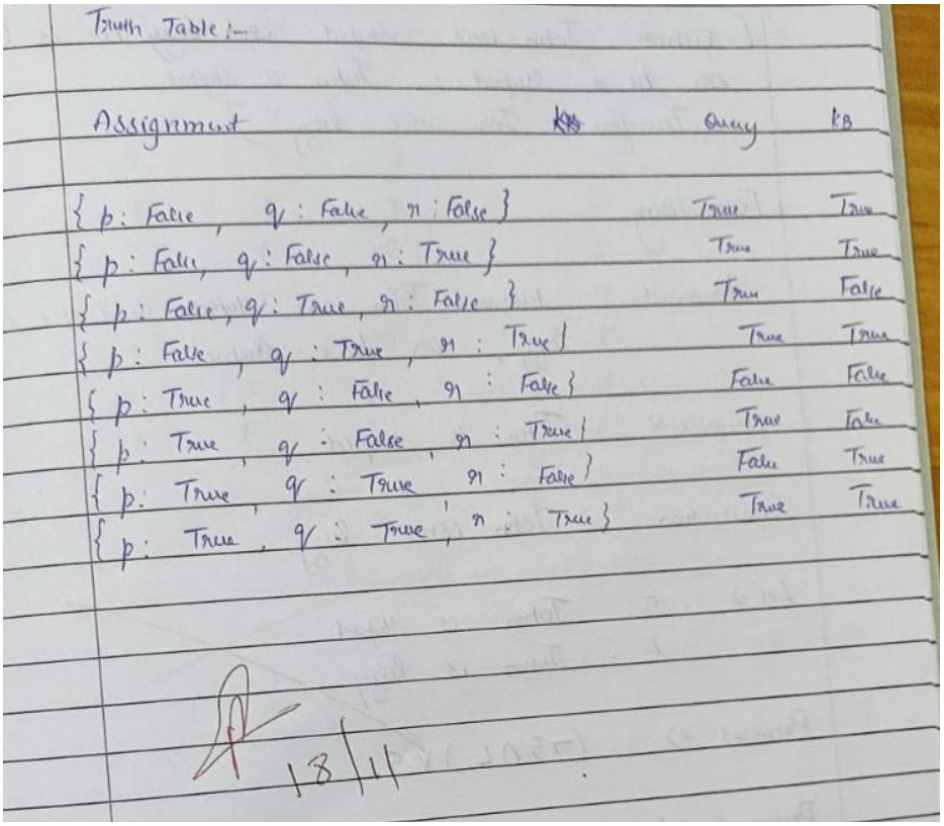
```
        return priority[c1] <= priority[c2]
    except KeyError:
        return False

def toPostfix(infix):
    stack = []
    postfix = ''
    for c in infix:
        if isOperand(c):
            postfix += c
        else:
            if isLeftParanthesis(c):
                stack.append(c)
            elif isRightParanthesis(c):
                operator = stack.pop()
                while not isLeftParanthesis(operator):
                    postfix += operator
                    operator = stack.pop()
            else:
                while (not isEmpty(stack)) and hasLessOrEqualPriority(c, peek(stack)):
                    postfix += stack.pop()
                stack.append(c)
    while (not isEmpty(stack)):
        postfix += stack.pop()
    return postfix

def evaluatePostfix(exp, comb):
    stack = []
    for i in exp:
        if isOperand(i):
            stack.append(comb[variable[i]])
        elif i == '~':
            val1 = stack.pop()
            stack.append(not val1)
        else:
            val1 = stack.pop()
            val2 = stack.pop()
            stack.append(_eval(i, val2, val1))
    return stack.pop()

def _eval(i, val1, val2):
    if i == '^':
        return val2 and val1
    return val2 or val1

input_rules()
```

```
ans = entailment()
if ans:
    print("The Knowledge Base entails query")
else:
    print("The Knowledge Base does not entail query")
```

OUTPUT:

```
Output

Initial configuration:
. . . Q
. Q . .
Q . . .
. . Q .

Initial number of conflicts: 1

Solution found at iteration 3!
Final configuration (solution):
. Q . .
. . . Q
Q . . .
. . Q .

=== Code Execution Successful ===
```

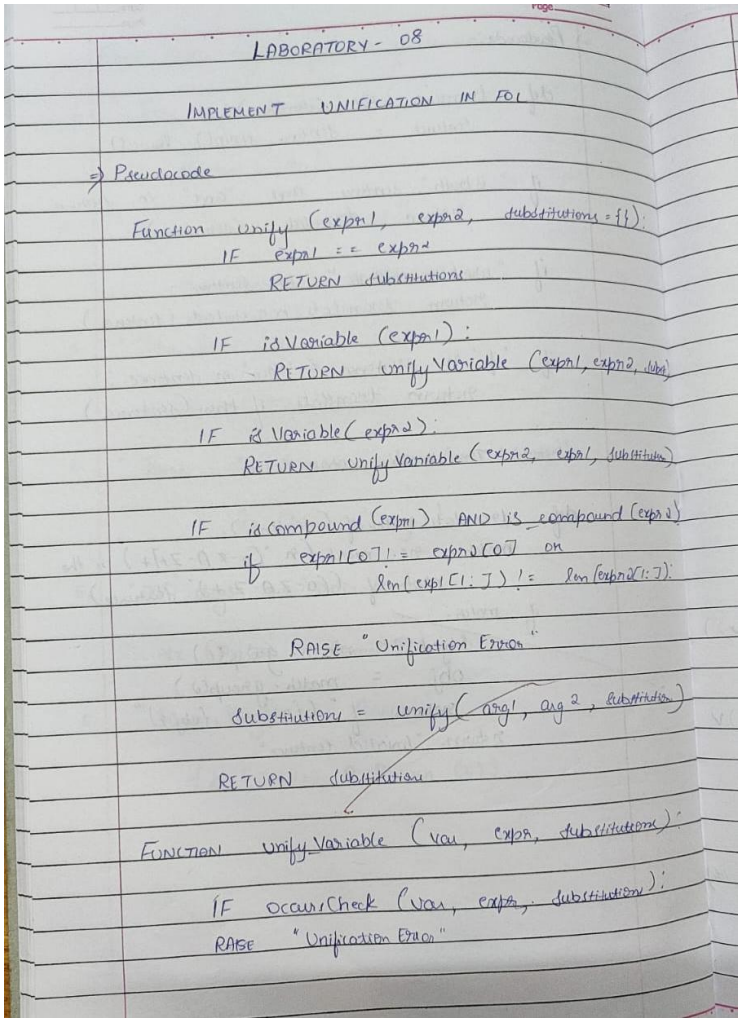
STATE SPACE TREE:



WEEK-7

Unification in First Order Logic

ALGORITHM:

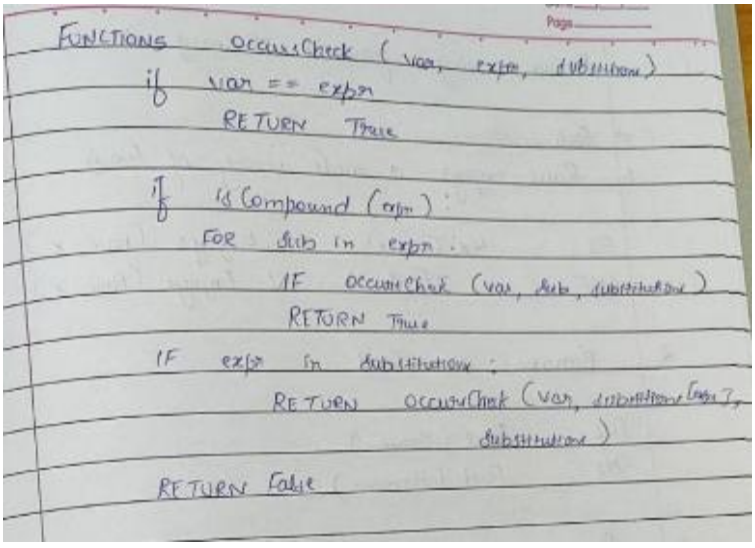


LABORATORY - 08

IMPLEMENT UNIFICATION IN FOL

=) Pseudocode

```
Function unify (expr1, expr2, substitutions={}):
    IF expr1 == expr2
        RETURN substitutions

    IF isVariable (expr1):
        RETURN unifyVariable (expr1, expr2, subs)

    IF is Variable(expr2):
        RETURN unifyVariable (expr2, expr1, substitution)

    IF isCompound (expr1) AND is_compound (expr2)
        if expr1[0] != expr2[0] or
            len(expr1[1:]) != len(expr2[1:]):

        RAISE "Unification Error"

    substitutions = unify( arg1, arg2, substitution)

    RETURN substitution

FUNCTION unifyVariable (var, expr, substitutions):

    IF occurCheck (var, expr, substitutions):
    RAISE "Unification Error"
```



```
FUNCTIONS occurCheck (var, expr, substitutions)
    if var == expr
        RETURN True

    if isCompound (expr)
        FOR sub in expr:
            IF occurCheck (var, sub, substitutions)
            RETURN True
    IF expr in substitutions:
        RETURN occurCheck (var, substitutions[expr],
                           substitutions)

    RETURN False
```

CODE:

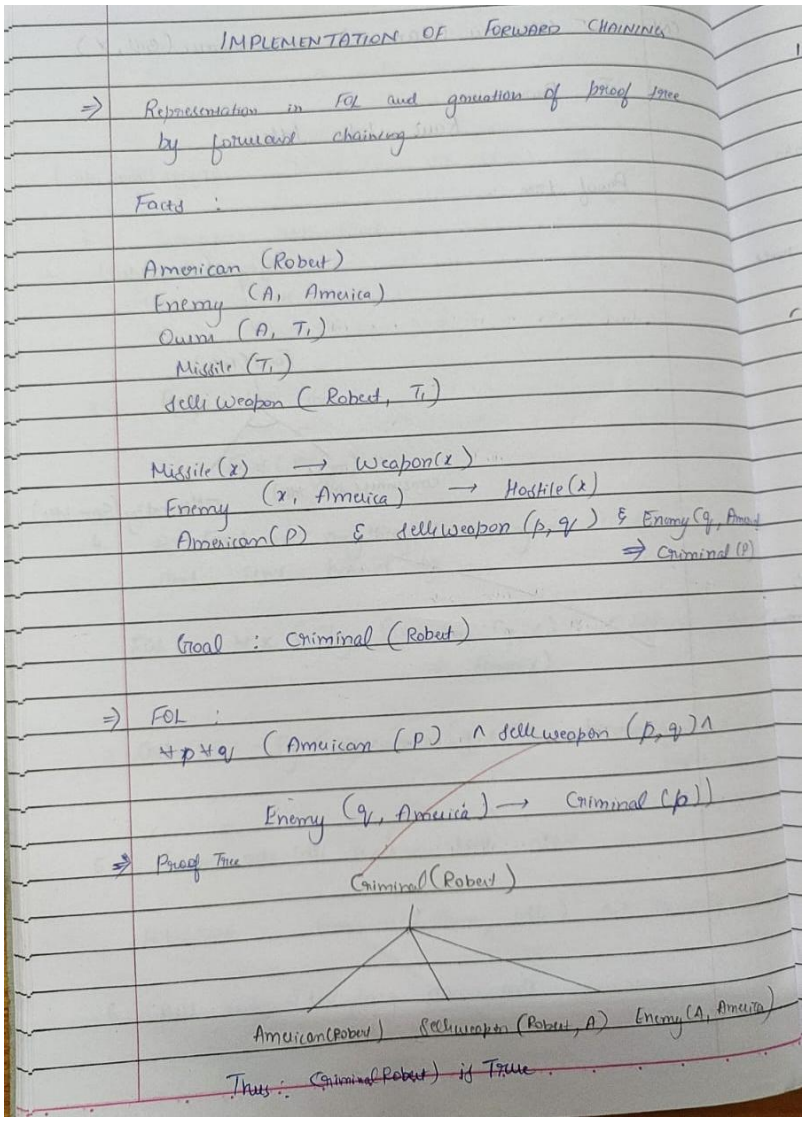
```
class UnificationError(Exception):
    pass

def unify(expr1, expr2, substitutions=None):
    if substitutions is None:
        substitutions = {}

    # If both expressions are identical, return current substitutions
    if expr1 == expr2:
        return substitutions

    # If the first expression is a variable
    if is_variable(expr1):
        return unify_variable(expr1, expr2, substitutions)

    # If the second expression is a variable
    if is_variable(expr2):
        return unify_variable(expr2, expr1, substitutions)

    # If both expressions are compound expressions
    if is_compound(expr1) and is_compound(expr2):
        if expr1[0] != expr2[0] or len(expr1[1:]) != len(expr2[1:]):
            raise UnificationError("Expressions do not match.")
        return unify_lists(expr1[1:], expr2[1:], unify(expr1[0], expr2[0],
substitutions))

    # If expressions are not compatible
    raise UnificationError(f"Cannot unify {expr1} and {expr2}.")

def unify_variable(var, expr, substitutions):
    if var in substitutions:
        return unify(substitutions[var], expr, substitutions)
    elif occurs_check(var, expr, substitutions):
        raise UnificationError(f"Occurs check failed: {var} in {expr}.")
    else:
        substitutions[var] = expr
        return substitutions
```

```
def unify_lists(list1, list2, substitutions):
    for expr1, expr2 in zip(list1, list2):
        substitutions = unify(expr1, expr2, substitutions)
    return substitutions

def is_variable(term):
    return isinstance(term, str) and term[0].islower()

def is_compound(term):
    return isinstance(term, (list, tuple)) and len(term) > 0

def occurs_check(var, expr, substitutions):
    if var == expr:
        return True
    elif is_compound(expr):
        return any(occurs_check(var, sub, substitutions) for sub in expr)
    elif expr in substitutions:
        return occurs_check(var, substitutions[expr], substitutions)
    return False

# Example usage
try:
    expr1 = ("f", "x", ("g", "y"))
    expr2 = ("f", "a", ("g", "b"))
    result = unify(expr1, expr2)
    print("Unified substitutions:", result)
except UnificationError as e:
    print("Unification failed:", e)
```

WEEK-8

10.Forward Chaining

ALGORITHM:



CODE:

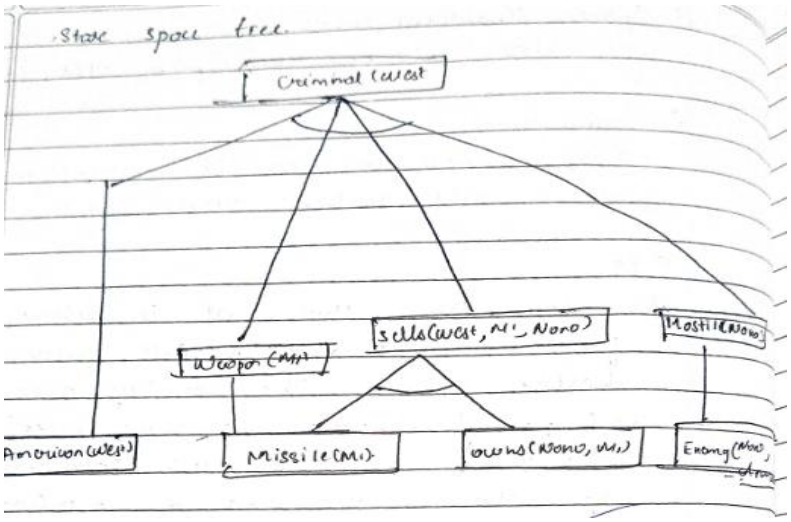
```
# Define initial facts and rules
facts = {"InAmerica(West)", "SoldWeapons(West, Nono)", "Enemy(Nono,
America)"}
rules = [
    {
        "conditions": ["InAmerica(x)", "SoldWeapons(x, y)", "Enemy(y, America)"],
        "conclusion": "Criminal(x)",
    },
    {
        "conditions": ["Enemy(y, America)"],
        "conclusion": "Dangerous(y)",
    },
]

# Forward chaining function
def forward_chaining(facts, rules):
    derived_facts = set(facts)  # Initialize derived facts
    while True:
        new_fact_found = False

        for rule in rules:
            # Substitute variables and check if conditions are met
            for fact in derived_facts:
                if "x" in rule["conditions"][0]:
                    # Substitute variables (x, y) with specific instances
                    for condition in rule["conditions"]:
                        if "x" in condition or "y" in condition:
                            x = "West"  # Hardcoded substitution for simplicity
                            y = "Nono"
                            conditions = [
                                cond.replace("x", x).replace("y", y)
                                for cond in rule["conditions"]
                            ]
                            conclusion = (
                                rule["conclusion"].replace("x", x).replace("y", y)
                            )
```

```
                    # Check if all conditions are satisfied
                    if all(cond in derived_facts for cond in conditions) and
conclusion not in derived_facts:
                        derived_facts.add(conclusion)
                        print(f"New fact derived: {conclusion}")
                        new_fact_found = True

        # Exit loop if no new fact is found
        if not new_fact_found:
            break

    return derived_facts

# Run forward chaining
final_facts = forward_chaining(facts, rules)
print("\nFinal derived facts:")
for fact in final_facts:
    print(fact)
```

OUTPUT :

```
Output

Final derived facts:
Enemy(Nono, America)
InAmerica(West)
SoldWeapons(West, Nono)

=== Code Execution Successful ===
```

STATE SPACE TREE:



WEEK-8

Resolution of First Order Logic statements

ALGORITHM:

CODE:

```
def negate(literal):
    """Return the negation of a literal."""
    if isinstance(literal, tuple) and literal[0] == "not":
        return literal[1]
    else:
        return ("not", literal)

def resolve(clause1, clause2):
    """Return the resolvent of two clauses."""
    resolvents = set()
    for literal1 in clause1:
        for literal2 in clause2:
            if literal1 == negate(literal2):
                resolvent = (clause1 - {literal1}) | (clause2 - {literal2})
                print(f"    Resolving literal: {literal1} with {literal2}")
                print(f"    Resulting Resolvent: {resolvent}")
                resolvents.add(frozenset(resolvent))
    return resolvents

def resolution_algorithm(KB, query):
    """Perform the resolution algorithm to check if the query can be proven."""
    print("\n--- Step-by-Step Resolution Process ---")
    negated_query = negate(query)
    KB.append(frozenset([negated_query]))
    print(f"Negated Query Added to KB: {negated_query}")

    clauses = set(KB)

    step = 1
    while True:
        new_clauses = set()
```

```
        print(f"\nStep {step}: Resolving Clauses")
        for c1 in clauses:
            for c2 in clauses:
                if c1 != c2:
                    print(f"  Resolving clauses: {c1} and {c2}")
                    resolvent = resolve(c1, c2)
                    for res in resolvent:
                        if frozenset([]) in resolvent:
                            print("\nEmpty clause derived! The query is provable.")
                            return True
                        new_clauses.add(res)

        if new_clauses.issubset(clauses):
            print("\nNo new clauses can be derived. The query is not provable.")
            return False

        clauses.update(new_clauses)
        step += 1

KB = [
    frozenset([("not", "food(x)"), ("likes", "John", "x")]),
    frozenset([("food", "Apple")]),
    frozenset([("food", "vegetables")]),
    frozenset([("not", "eats(y, z)"), ("killed", "y"), ("food", "z")]),
    frozenset([("eats", "Anil", "Peanuts")]),
    frozenset([("alive", "Anil")]),
    frozenset([("not", "eats(Anil, w)"), ("eats", "Harry", "w")]),
    frozenset([("killed", "g"), ("alive", "g")]),
    frozenset([("not", "alive(k)"), ("not", "killed(k)")]),
    frozenset([("likes", "John", "Peanuts")])
]


query = ("likes", "John", "Peanuts")

result = resolution_algorithm(KB, query)
if result:
    print("\nQuery is provable.")
else:
    print("\nQuery is not provable.")
```

OUTPUT :

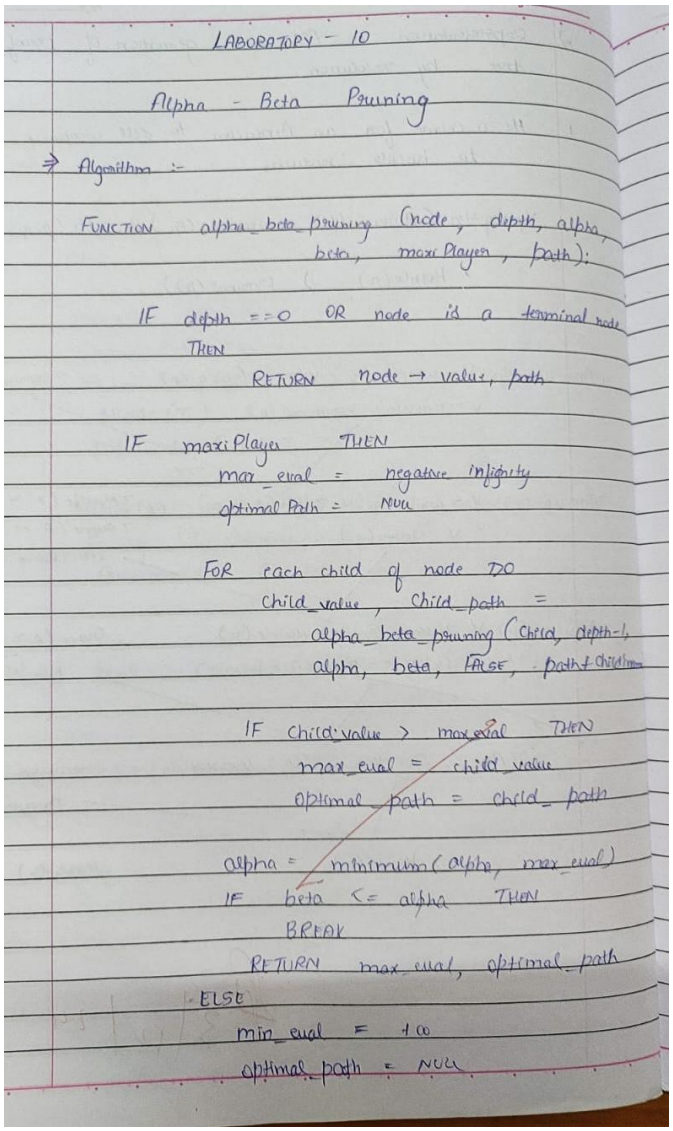
```
  Resolving clauses: frozenset({('food', 'Apple')}) and frozenset({('alive', 'g'), ('killed', 'g')})
  Resolving clauses: frozenset({('food', 'Apple')}) and frozenset({('eats', 'Harry', 'w'), ('not', 'eats(Anil, w)')})
  Resolving clauses: frozenset({('food', 'Apple')}) and frozenset({('not', 'killed(k)'), ('not', 'alive(k)')})
  Resolving clauses: frozenset({('food', 'Apple')}) and frozenset({('eats', 'Anil', 'Peanuts')})
  Resolving clauses: frozenset({('food', 'Apple')}) and frozenset({('not', ('likes', 'John', 'Peanuts'))})
  Resolving clauses: frozenset({('food', 'Apple')}) and frozenset({('food', 'vegetables')})
  Resolving clauses: frozenset({('not', 'food(x)'), ('likes', 'John', 'x')}) and frozenset({('food', 'Apple')})
  Resolving clauses: frozenset({('not', 'food(x)'), ('likes', 'John', 'x')}) and frozenset({('killed', 'y'), ('not', 'eats(y, z)'), ('food', 'z')})
  Resolving clauses: frozenset({('not', 'food(x)'), ('likes', 'John', 'x')}) and frozenset({('likes', 'John', 'Peanuts')})
  Resolving clauses: frozenset({('not', 'food(x)'), ('likes', 'John', 'x')}) and frozenset({('alive', 'Anil')})
  Resolving clauses: frozenset({('not', 'food(x)'), ('likes', 'John', 'x')}) and frozenset({('alive', 'g'), ('killed', 'g')})
  Resolving clauses: frozenset({('not', 'food(x)'), ('likes', 'John', 'x')}) and frozenset({('eats', 'Harry', 'w'), ('not', 'eats(Anil, w)')})
  Resolving clauses: frozenset({('not', 'food(x)'), ('likes', 'John', 'x')}) and frozenset({('not', 'killed(k)'), ('not', 'alive(k)')})
  Resolving clauses: frozenset({('not', 'food(x)'), ('likes', 'John', 'x')}) and frozenset({('eats', 'Anil', 'Peanuts')})
  Resolving clauses: frozenset({('not', 'food(x)'), ('likes', 'John', 'x')}) and frozenset({('not', ('likes', 'John', 'Peanuts'))})
  Resolving clauses: frozenset({('not', 'food(x)'), ('likes', 'John', 'x')}) and frozenset({('food', 'vegetables')})
  Resolving clauses: frozenset({('killed', 'y'), ('not', 'eats(y, z)'), ('food', 'z')}) and frozenset({('food', 'Apple')})
  Resolving clauses: frozenset({('killed', 'y'), ('not', 'eats(y, z)'), ('food', 'z')}) and frozenset({('not', 'food(x)'), ('likes', 'John', 'x')})
  Resolving clauses: frozenset({('killed', 'y'), ('not', 'eats(y, z)'), ('food', 'z')}) and frozenset({('likes', 'John', 'Peanuts')})
  Resolving clauses: frozenset({('killed', 'y'), ('not', 'eats(y, z)'), ('food', 'z')}) and frozenset({('alive', 'Anil')})
  Resolving clauses: frozenset({('killed', 'y'), ('not', 'eats(y, z)'), ('food', 'z')}) and frozenset({('alive', 'g'), ('killed', 'g')})
  Resolving clauses: frozenset({('killed', 'y'), ('not', 'eats(y, z)'), ('food', 'z')}) and frozenset({('eats', 'Harry', 'w'), ('not', 'eats(Anil, w)')})
  Resolving clauses: frozenset({('killed', 'y'), ('not', 'eats(y, z)'), ('food', 'z')}) and frozenset({('not', 'killed(k)'), ('not', 'alive(k)')})
  Resolving clauses: frozenset({('killed', 'y'), ('not', 'eats(y, z)'), ('food', 'z')}) and frozenset({('eats', 'Anil', 'Peanuts')})
  Resolving clauses: frozenset({('killed', 'y'), ('not', 'eats(y, z)'), ('food', 'z')}) and frozenset({('not', ('likes', 'John', 'Peanuts'))})
  Resolving clauses: frozenset({('killed', 'y'), ('not', 'eats(y, z)'), ('food', 'z')}) and frozenset({('food', 'vegetables')})
  Resolving clauses: frozenset({('likes', 'John', 'Peanuts')}) and frozenset({('food', 'Apple')})
  Resolving clauses: frozenset({('likes', 'John', 'Peanuts')}) and frozenset({('not', 'food(x)'), ('likes', 'John', 'x')})
  Resolving clauses: frozenset({('likes', 'John', 'Peanuts')}) and frozenset({('killed', 'y'), ('not', 'eats(y, z)'), ('food', 'z')})
  Resolving clauses: frozenset({('likes', 'John', 'Peanuts')}) and frozenset({('alive', 'Anil')})
  Resolving clauses: frozenset({('likes', 'John', 'Peanuts')}) and frozenset({('alive', 'g'), ('killed', 'g')})
  Resolving clauses: frozenset({('likes', 'John', 'Peanuts')}) and frozenset({('eats', 'Harry', 'w'), ('not', 'eats(Anil, w)')})
  Resolving clauses: frozenset({('likes', 'John', 'Peanuts')}) and frozenset({('not', 'killed(k)'), ('not', 'alive(k)')})
  Resolving clauses: frozenset({('likes', 'John', 'Peanuts')}) and frozenset({('eats', 'Anil', 'Peanuts')})
  Resolving clauses: frozenset({('likes', 'John', 'Peanuts')}) and frozenset({('not', ('likes', 'John', 'Peanuts'))})
    Resolving literal: ('likes', 'John', 'Peanuts') with ('not', ('likes', 'John', 'Peanuts'))
    Resulting Resolvent: frozenset()

Empty clause derived! The query is provable.
```

TRACING :

WEEK-9

Alpha-Beta Pruning

ALGORITHM :



LABORATORY - 10

Alpha - Beta Pruning

⇒ Algorithm :-

FUNCTION alpha_beta_pruning (node, depth, alpha, beta, maxiPlayer, path):

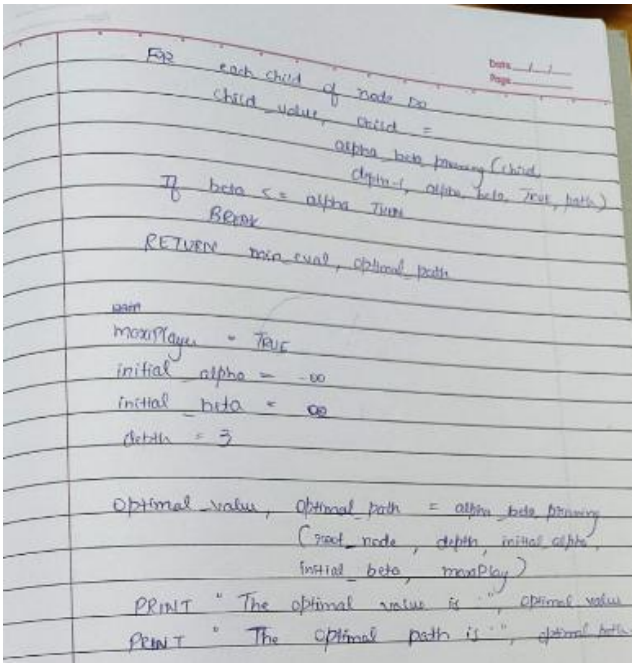
IF depth == 0 OR node is a terminal node THEN
    RETURN node → value, path

IF maxiPlayer THEN
    max_eval = negative infinity
    optimal Path = NULL

    FOR each child of node DO
        child_value, child_path = alpha_beta_pruning (child, depth-1, alpha, beta, FALSE, path+child)

        IF child_value > max_eval THEN
            max_eval = child_value
            optimal_path = child_path

        alpha = minimum (alpha, max_eval)
        IF beta <= alpha THEN
            BREAK
    RETURN max_eval, optimal_path
ELSE
    min_eval = +∞
    optimal_path = NULL



FOR each child of node DO
    child_value, child = alpha_beta_pruning (child, depth-1, alpha, beta, True, path)
    If beta <= alpha THEN
        BREAK
RETURN min_eval, optimal_path

main
maxiPlayer = TRUE
initial_alpha = -∞
initial_beta = ∞
depth = 3

Optimal_value, optimal_path = alpha_beta_pruning (root_node, depth, initial_alpha, initial_beta, maxiPlay)
PRINT "The optimal value is :", optimal_value
PRINT "The optimal path is :", optimal_path

CODE:

```
def alpha_beta_search(state):

    def max_value(state, alpha, beta, path):
```

```
    print(f"MAX: Visiting state {state}, alpha={alpha}, beta={beta}")
    if terminal_test(state):
        print(f"MAX: Terminal state {state} has utility {utility(state)}")
        return utility(state), path
    v = float('-inf')
    best_path = []
    for action in actions(state):
        result_state = result(state, action)
        value, new_path = min_value(result_state, alpha, beta, path + [action])
        print(f"MAX: From state {state}, action {action} → value={value}")
        if value > v:
            v = value
            best_path = new_path
        if v >= beta:
            print(f"MAX: Pruning at state {state} with value={v} ≥ beta={beta}")
            return v, best_path
        alpha = max(alpha, v)
    print(f"MAX: Returning value={v} for state {state}")
    return v, best_path

def min_value(state, alpha, beta, path):
    print(f"MIN: Visiting state {state}, alpha={alpha}, beta={beta}")
    if terminal_test(state):
        print(f"MIN: Terminal state {state} has utility {utility(state)}")
        return utility(state), path
```

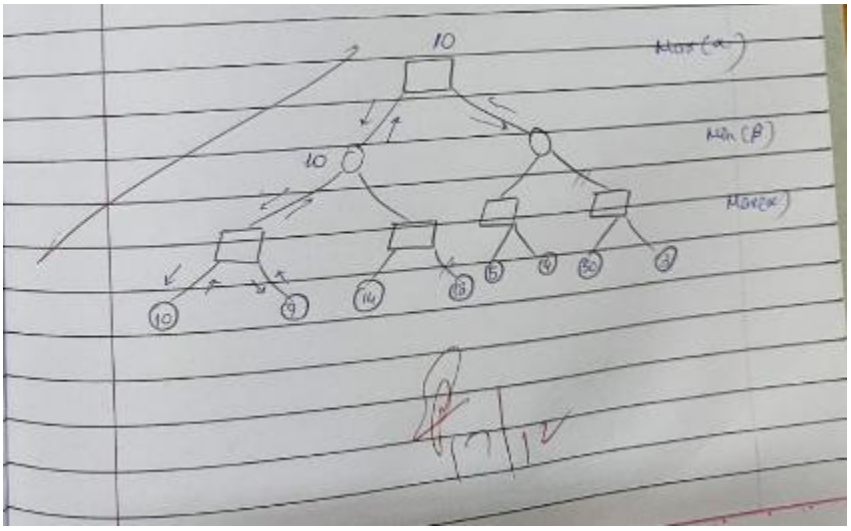
```
        v = float('inf')
        best_path = []
        for action in actions(state):
            result_state = result(state, action)
            value, new_path = max_value(result_state, alpha, beta, path + [action])
            print(f"MIN: From state {state}, action {action} → value={value}")
            if value < v:
                v = value
                best_path = new_path
            if v <= alpha:
                print(f"MIN: Pruning at state {state} with value={v} ≤ alpha={alpha}")
                return v, best_path
            beta = min(beta, v)
        print(f"MIN: Returning value={v} for state {state}")
        return v, best_path

    print("Starting Alpha-Beta Search...\n")
    final_value, final_path = max_value(state, float('-inf'), float('inf'), [state])
    return final_value, final_path


def terminal_test(state):
    return state in terminal_states


def utility(state):
```

```
    return terminal_states[state]


def actions(state):
    return game_tree.get(state, [])
```

OUTPUT :

```
Starting Alpha-Beta Search...

MAX: Visiting state A, alpha=-inf, beta=inf
MIN: Visiting state B, alpha=-inf, beta=inf
MAX: Visiting state D, alpha=-inf, beta=inf
MIN: Visiting state H, alpha=-inf, beta=inf
MIN: Terminal state H has utility 10
MAX: From state D, action H → value=10
MIN: Visiting state I, alpha=10, beta=inf
MIN: Terminal state I has utility 9
MAX: From state D, action I → value=9
MAX: Returning value=10 for state D
MIN: From state B, action D → value=10
MAX: Visiting state E, alpha=-inf, beta=10
MIN: Visiting state J, alpha=-inf, beta=10
MIN: Terminal state J has utility 14
MAX: From state E, action J → value=14
MAX: Pruning at state E with value=14 ≥ beta=10
MIN: From state B, action E → value=14
MIN: Returning value=10 for state B
MAX: From state A, action B → value=10
MIN: Visiting state C, alpha=10, beta=inf
MAX: Visiting state F, alpha=10, beta=inf
MIN: Visiting state L, alpha=10, beta=inf
MIN: Terminal state L has utility 5
MAX: From state F, action L → value=5
MIN: Visiting state M, alpha=10, beta=inf
MIN: Terminal state M has utility 4
MAX: From state F, action M → value=4
MAX: Returning value=5 for state F
MIN: From state C, action F → value=5
MIN: Pruning at state C with value=5 ≤ alpha=10
MAX: From state A, action C → value=5
MAX: Returning value=10 for state A

--- Final Results ---
The final utility value is: 10
The final path taken is: ['A', 'B', 'D', 'H']
```

STATE SPACE TREE :


10
Max(α)
10
Min(β)
Max(α)
10
9
14
18
15
4
80
2